News Impact Modeling Using a Two-Stage XGBoost Architecture for Stock Forecasting

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DECLARATION OF THE CANDIDATE AND SUPERVISOR

I declare that this is my own work and this proposal does not incorporate without acknowledgment any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgment is made in the text.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

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Signature of the co-supervisor: Date

Abstract

This study presents a sentiment analysis framework designed to extract and quantify market-relevant information from financial textual data. In modern financial ecosystems, unstructured data sources such as news articles, corporate disclosures, and financial reports significantly influence investor behavior and market movements. The objective of this research is to convert such qualitative information into structured sentiment indicators that can provide actionable insights. The proposed methodology involves systematic data collection followed by preprocessing steps, including tokenization, stop-word removal, and text normalization. Natural Language Processing (NLP) techniques are applied to transform textual content into numerical representations. A hybrid sentiment classification approach is implemented, combining lexicon-based methods with supervised machine learning algorithms to categorize text into positive, negative, or neutral sentiment classes. The model is trained and validated using labeled datasets to ensure reliability and generalization. Experimental results demonstrate that the developed framework effectively captures underlying sentiment patterns within financial text and identifies shifts in market perception. The findings indicate that sentiment analysis can serve as a reliable tool for interpreting qualitative financial information and enhancing decision-making processes. In conclusion, this research highlights the significance of sentiment analysis in financial text mining and its potential to generate meaningful market insights. Future work should explore the integration of advanced deep learning techniques and real-time data streams to further improve accuracy and responsiveness.

Keywords: ***Sentiment analysis, financial text mining, natural language processing, machine learning, market sentiment***

Acknowledgement

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1. INTRODUCTION

Financial markets operate as complex, information-driven systems in which prices are continuously influenced by the flow and interpretation of new information. Traditionally, financial analysis has focused on structured quantitative data such as historical prices, financial ratios, earnings reports, and macroeconomic indicators. While these factors remain fundamental, they do not fully capture the behavioral and psychological elements that significantly affect market dynamics. A growing body of evidence suggests that investor sentiment—shaped by opinions, expectations, and reactions to news—plays a critical role in explaining short-term market movements and volatility. Much of this sentiment is embedded in unstructured textual data, including financial news articles, corporate announcements, earnings call transcripts, and social media discussions.

With the rapid expansion of digital information sources, the volume of financial text data has increased exponentially, making manual analysis impractical and inefficient. As a result, automated techniques for extracting meaningful insights from textual data have become essential. Sentiment analysis, a specialized area within Natural Language Processing (NLP), offers a systematic approach to identifying and quantifying subjective information in text. By classifying textual content into sentiment categories such as positive, negative, or neutral, sentiment analysis enables the transformation of qualitative narratives into structured, machine-readable features.

In the context of financial markets, sentiment analysis serves as a bridge between qualitative information and quantitative modeling. It allows analysts to capture market perception, investor confidence, and reaction to events, which are often not directly observable through numerical data alone. For instance, a company's financial results may meet expectations, yet negative sentiment in news coverage can still lead to a decline in stock prices. Conversely, optimistic sentiment can drive price increases even in the absence of strong fundamentals. Therefore, incorporating sentiment-based insights can enhance the understanding of market behavior and improve decision-making processes.

Despite its potential, financial sentiment analysis presents several domain-specific challenges. Financial text is characterized by specialized vocabulary, complex sentence structures, and context-dependent meanings. Words that are typically associated with negative sentiment in general language may carry neutral or even positive connotations in financial contexts. For example, terms such as "liability" or "debt" are not inherently negative within financial reporting. Additionally, financial narratives often include nuanced expressions, forward-looking statements, and implicit signals that require deeper contextual understanding. These complexities limit the effectiveness of generic sentiment analysis models and necessitate tailored approaches designed specifically for financial data.

To address these challenges, this research proposes a comprehensive sentiment analysis framework for financial textual data. The approach begins with the systematic collection of

relevant financial text from reliable sources, ensuring that the dataset reflects diverse market conditions and information types. The collected data undergoes rigorous preprocessing, including tokenization, stop-word removal, normalization, and, where necessary, lemmatization or stemming to standardize textual inputs. Feature extraction techniques such as bag-of-words and term frequency–inverse document frequency (TF-IDF) are employed to convert textual data into numerical representations suitable for machine learning models.

A hybrid sentiment classification strategy is adopted to improve accuracy and robustness. Lexicon-based methods are utilized to provide an interpretable baseline by assigning sentiment scores based on predefined dictionaries of financial terms. These are complemented by supervised machine learning algorithms, which learn patterns from labeled data and adapt to domain-specific language usage. The combination of these approaches aims to balance interpretability with predictive performance, overcoming the limitations associated with relying on a single method.

The system is evaluated using appropriate performance metrics, including accuracy, precision, recall, and F1-score, to assess its classification effectiveness. Experimental results demonstrate the model's capability to capture sentiment trends and distinguish between different sentiment classes with a satisfactory level of reliability. The findings highlight the importance of domain adaptation and the integration of multiple techniques in achieving meaningful results in financial text analysis.

The significance of this study lies in its contribution to the growing field of financial text mining and its practical implications for market participants. By converting unstructured textual information into actionable sentiment indicators, the proposed framework can support investors, analysts, and financial institutions in making more informed decisions. It also opens avenues for further integration with predictive models, risk assessment systems, and real-time market monitoring tools.

The remainder of this paper is structured as follows: Section II reviews existing literature related to sentiment analysis and financial text mining, highlighting key methodologies and limitations. Section III details the proposed methodology, including data collection, preprocessing, feature extraction, and model development. Section IV presents the experimental setup, results, and discussion. Finally, Section V concludes the study and outlines potential directions for future research, including the application of deep learning models and real-time sentiment analysis systems.

1.1 Background & Literature Survey

The analysis of financial markets has traditionally been grounded in structured, numerical data such as stock prices, financial ratios, and macroeconomic indicators. However, with the rapid digitization of information and the proliferation of online content, unstructured textual data has emerged as a critical source of market-relevant information. Financial news articles, corporate disclosures, earnings call transcripts, analyst reports, and even social media discussions continuously shape investor perception and influence market behavior. As a result, there has been a growing need for computational techniques capable of extracting meaningful insights from such data. Sentiment analysis, a specialized domain within Natural Language Processing (NLP), addresses this need by enabling the identification and quantification of subjective information embedded within text.

A. Foundations of Sentiment Analysis

Sentiment analysis, also referred to as opinion mining, involves the use of computational methods to determine the polarity of text—typically categorized as positive, negative, or neutral. Early developments in this field were primarily driven by applications in consumer analytics, where businesses sought to analyze customer opinions from product reviews and social media platforms. These initial approaches laid the groundwork for subsequent adaptations in more complex domains such as finance.

At its core, sentiment analysis involves several key steps: text preprocessing, feature extraction, and sentiment classification. Preprocessing typically includes tokenization, stop-word removal, normalization, and stemming or lemmatization. Feature extraction methods, such as bag-of-words and term frequency–inverse document frequency (TF-IDF), convert textual data into numerical representations. These representations are then used as input for classification algorithms that determine the sentiment polarity of the text.

While these general methodologies provide a strong foundation, their direct application to financial text is not straightforward. Financial language exhibits unique characteristics, including specialized vocabulary, context-dependent meanings, and a high degree of implicit information. Consequently, domain adaptation is essential for achieving meaningful results in financial sentiment analysis.

B. Lexicon-Based Approaches in Financial Sentiment Analysis

Lexicon-based methods represent one of the earliest approaches to sentiment analysis. These techniques rely on predefined dictionaries of words, where each word is assigned a sentiment score. The overall sentiment of a document is computed by aggregating the scores of individual words or phrases. The primary advantage of lexicon-based methods lies in their simplicity and interpretability, as the sentiment classification process is transparent and easily explainable.

In the financial domain, researchers recognized that general-purpose sentiment lexicons were insufficient due to domain-specific language usage. Words that carry negative connotations in everyday language may be neutral or even positive in financial contexts. For example, terms such as “liability,” “depreciation,” or “risk” are not inherently negative within financial reporting. To address this issue, domain-specific lexicons were developed, incorporating financial terminology and context-aware sentiment scores.

Empirical studies have demonstrated that financial lexicons significantly improve classification accuracy compared to generic dictionaries. However, lexicon-based methods are inherently limited in their ability to capture contextual nuances, syntactic relationships, and semantic dependencies. They often fail to account for negation, sarcasm, and complex sentence structures, which are common in financial narratives. As a result, while lexicon-based approaches provide a useful baseline, they are generally insufficient for achieving high-performance sentiment classification on their own.

C. Machine Learning Approaches

The limitations of lexicon-based methods led to the adoption of machine learning techniques for sentiment analysis. Supervised learning algorithms such as Naïve Bayes, Support Vector Machines (SVM), Decision Trees, and Logistic Regression became widely used due to their ability to learn patterns from labeled data. These models rely on feature extraction techniques to represent text numerically, enabling them to identify relationships between textual features and sentiment labels.

Machine learning approaches offer several advantages over lexicon-based methods. They can capture complex patterns in data, adapt to domain-specific language, and improve performance with larger datasets. In the context of financial sentiment analysis, studies have shown that machine learning models outperform lexicon-based approaches in terms of accuracy and robustness. For example, SVM models have been particularly effective due to their ability to handle high-dimensional feature spaces and sparse data.

Despite these advantages, machine learning approaches also present challenges. Their performance is highly dependent on the availability and quality of labeled datasets, which can be costly and time-consuming to obtain. Additionally, traditional machine learning models often require extensive feature engineering, which may not fully capture the semantic richness of financial text. Overfitting and model interpretability are also important considerations, particularly in financial applications where transparency is critical.

D. Deep Learning and Advanced NLP Techniques

Recent advancements in deep learning have significantly transformed the field of sentiment analysis. Neural network architectures such as Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM) networks, and Convolutional Neural Networks (CNNs) have demonstrated strong performance in capturing sequential and contextual information in text. These models are capable of learning hierarchical representations of language, reducing the need for manual feature engineering.

In financial sentiment analysis, deep learning models have been applied to various tasks, including sentiment classification of news articles, prediction of stock price movements, and analysis of earnings call transcripts. LSTM networks, in particular, have proven effective in modeling temporal dependencies and long-range context within financial text. More recently, Transformer-based models such as BERT (Bidirectional Encoder Representations from Transformers) have achieved state-of-the-art performance in many NLP tasks, including sentiment analysis. These models leverage attention mechanisms to capture complex relationships between words, enabling a deeper understanding of textual context.

However, the application of deep learning in financial sentiment analysis is not without challenges. These models require large volumes of labeled data and significant computational resources for training. Additionally, their complexity can make them less interpretable compared to traditional methods, which may limit their adoption in certain financial applications where explainability is essential.

E. Sentiment Analysis in Financial Markets

A substantial body of literature has explored the relationship between textual sentiment and financial market behavior. Studies have shown that sentiment derived from news articles, corporate disclosures, and social media can provide valuable predictive signals for stock price movements, volatility, and trading volume. For instance, positive sentiment in earnings announcements has been associated with abnormal returns, while negative sentiment often correlates with increased market volatility.

The integration of sentiment analysis into financial models has been shown to enhance predictive performance. By combining sentiment indicators with traditional financial metrics, researchers have developed hybrid models that capture both quantitative and qualitative aspects of market behavior. These models have been applied in various contexts, including algorithmic trading, risk assessment, and portfolio management.

Social media platforms, in particular, have emerged as a rich source of real-time sentiment data. Platforms such as Twitter and financial forums provide insights into investor opinions and market sentiment at a granular level. However, the use of social media data introduces additional challenges, including noise, misinformation, and the need for robust filtering mechanisms.

F. Hybrid Approaches and Emerging Trends

To overcome the limitations of individual methods, recent research has increasingly focused on hybrid approaches that combine lexicon-based techniques with machine learning or deep learning models. These approaches aim to leverage the strengths of each method, achieving a balance between interpretability and predictive performance. For example, lexicon-based features can be used as inputs to machine learning models, enhancing their ability to capture domain-specific sentiment signals.

Another emerging trend is the use of real-time sentiment analysis systems, which process streaming data to provide up-to-date insights into market sentiment. These systems are

particularly relevant in high-frequency trading environments, where timely information is critical. Additionally, there is growing interest in multimodal analysis, where textual sentiment is combined with other data sources such as images, audio, and numerical data to provide a more comprehensive view of market behavior.

G. Research Gaps and Motivation

Despite significant advancements, several challenges remain in financial sentiment analysis. Domain specificity, data scarcity, and the dynamic nature of financial markets continue to pose difficulties for model development and evaluation. Furthermore, achieving a balance between model accuracy and interpretability remains an ongoing concern.

This study is motivated by the need to address these challenges through the development of a hybrid sentiment analysis framework tailored to financial textual data. By integrating lexicon-based methods with machine learning techniques, the proposed approach aims to achieve robust classification performance while maintaining interpretability. The research also seeks to contribute to the broader field of financial text mining by providing a structured methodology for extracting and analyzing sentiment from unstructured data.

H. Summary

In summary, the literature reveals a clear progression in sentiment analysis techniques, from simple lexicon-based methods to advanced machine learning and deep learning approaches. While each method offers distinct advantages, no single approach fully addresses the complexities of financial text. Hybrid methodologies and domain-specific adaptations have emerged as promising solutions, highlighting the importance of continued research in this area. This study builds upon these developments, aiming to provide an effective and scalable sentiment analysis framework for financial applications.

1.2 Research gap

Although sentiment analysis has matured significantly as a research domain, its application to financial textual data still exhibits several unresolved challenges that limit both theoretical advancement and practical deployment. Beyond the commonly identified issues, deeper examination of the literature reveals structural, methodological, and operational gaps that justify the need for more robust and domain-aware solutions.

A critical gap exists in the **temporal modeling of sentiment dynamics**. Most existing studies treat sentiment as a static attribute, assigning a polarity score to a document without considering how sentiment evolves over time. In financial markets, however, the timing, persistence, and transition of sentiment are crucial. For instance, a sudden shift from positive to negative sentiment may have a stronger market impact than consistently neutral sentiment. Current models often fail to capture such temporal dependencies, limiting their ability to reflect real market behavior.

Another underexplored area is the **granularity of sentiment analysis**. Many approaches operate at the document or sentence level, which can obscure fine-grained sentiment signals embedded within specific phrases or entities. Financial text often contains mixed sentiment—for example, a report may present strong revenue growth alongside concerns about future risks. Existing models may oversimplify such cases by assigning a single sentiment label, thereby losing valuable information. There is a need for more granular approaches, such as aspect-based or entity-level sentiment analysis, specifically tailored to financial contexts.

The **lack of standardized benchmarks and evaluation frameworks** also represents a significant research gap. Different studies use varying datasets, preprocessing techniques, and evaluation metrics, making it difficult to compare results or establish best practices. Moreover, many evaluations are conducted in controlled experimental settings that do not accurately reflect real-world conditions. This lack of standardization hinders the reproducibility and scalability of sentiment analysis models in financial applications.

Another important limitation is the **insufficient handling of noisy and heterogeneous data sources**. Financial text data originates from diverse channels, including formal reports, news media, and informal social platforms. Each source exhibits different linguistic characteristics, levels of reliability, and noise patterns. Existing models often focus on a single data source, limiting their generalizability. There is a gap in developing unified frameworks capable of effectively integrating and processing multi-source financial text data while filtering noise and misinformation.

The issue of **data imbalance and bias** is also prevalent in financial sentiment datasets. Positive and neutral sentiments often dominate, while negative sentiments may be underrepresented or contextually subtle. This imbalance can lead to biased models that perform poorly in detecting critical negative signals, which are often the most impactful in financial decision-making. Additionally, biases in data collection—such as overrepresentation of certain sectors or companies—can further distort model performance.

From a methodological perspective, there is a gap in the **integration of linguistic features with domain knowledge**. Many models rely heavily on statistical or neural representations of text, with limited incorporation of financial theories or expert knowledge. For example, macroeconomic indicators, industry-specific trends, and corporate governance factors are rarely integrated into sentiment analysis frameworks. Bridging this gap could lead to more context-aware and economically meaningful interpretations of sentiment.

Another emerging concern is the **interpretability of complex models**, particularly deep learning architectures. While models such as Transformers achieve high accuracy, they often function as "black boxes," making it difficult to understand how specific predictions are derived. In financial applications, where decisions may involve significant risk and regulatory scrutiny, the lack of explainability poses a major barrier to adoption. There is a clear need for approaches that provide both high performance and transparent decision-making processes.

The **adaptability of sentiment models to changing market conditions** is also insufficiently addressed. Financial markets are dynamic, with evolving language patterns, emerging terminology, and shifting investor behavior. Models trained on historical data may become outdated if they cannot adapt to new contexts. Current research offers limited solutions for continuous learning or model updating mechanisms that maintain relevance over time.

Furthermore, **real-time deployment and scalability** remain practical challenges. While many studies demonstrate promising results in offline settings, fewer address the computational and architectural requirements for deploying sentiment analysis systems in real-time environments. Issues such as latency, data throughput, and system integration are often overlooked, despite their importance in applications like algorithmic trading and live market monitoring.

Another gap lies in the **limited exploration of cross-market and cross-lingual sentiment analysis**. Most research focuses on English-language datasets and specific markets, which restricts the applicability of findings to global financial systems. In multilingual and emerging markets, such as those with diverse linguistic and economic contexts, sentiment analysis models must account for language variations and localized financial expressions.

Finally, there is a notable lack of **end-to-end frameworks that connect sentiment extraction to actionable financial outcomes**. While many studies successfully classify sentiment, fewer demonstrate how these outputs can be directly integrated into decision-making systems or predictive models. Bridging this gap requires not only accurate sentiment classification but also effective mapping of sentiment signals to financial metrics such as price movements, volatility, or risk indicators.

1.3. Research problem

Despite the growing adoption of sentiment analysis in financial domains, there remains a fundamental challenge in effectively extracting, interpreting, and utilizing sentiment from unstructured financial textual data in a manner that is both accurate and practically applicable. Financial markets are highly sensitive to information, yet a significant portion of this information

exists in textual form, making it difficult to systematically incorporate into analytical and decision-making frameworks.

The core research problem addressed in this study is the **lack of a robust, domain-adapted sentiment analysis framework capable of accurately capturing the nuanced, context-dependent nature of financial text while maintaining interpretability and practical usability**. Existing approaches exhibit notable limitations that prevent them from fully addressing this problem.

Firstly, lexicon-based methods, while interpretable, fail to capture contextual meaning and often misclassify financial terminology due to their reliance on static dictionaries. Secondly, machine learning models improve classification performance but depend heavily on labeled datasets and require extensive feature engineering, which may not generalize well across different financial contexts. Thirdly, deep learning approaches offer superior contextual understanding but introduce challenges related to computational complexity, data requirements, and lack of transparency.

In addition, financial textual data presents unique characteristics that further complicate sentiment analysis. These include domain-specific vocabulary, implicit meanings, mixed sentiment within single documents, and temporal variations in sentiment patterns. Existing models often treat sentiment as static and fail to account for its dynamic evolution over time. Moreover, the diversity of data sources—ranging from formal reports to informal social media—introduces variability and noise that are not adequately handled by conventional methods.

Another dimension of the research problem is the **difficulty in translating extracted sentiment into meaningful and actionable insights**. While many studies focus on improving sentiment classification accuracy, fewer address how these sentiment outputs can be effectively utilized in real-world financial applications. This creates a gap between theoretical model performance and practical utility.

Therefore, the problem can be formally stated as follows:
How can a sentiment analysis system be designed to accurately and efficiently extract, interpret, and quantify sentiment from financial textual data, while addressing domain-specific challenges, ensuring interpretability, and enabling meaningful application in financial decision-making?

To address this problem, this research aims to develop a hybrid sentiment analysis framework that integrates lexicon-based and machine learning techniques. The proposed solution seeks to balance accuracy and interpretability, adapt to domain-specific language, and provide a structured approach for converting unstructured financial text into actionable sentiment indicators.

Ultimately, solving this research problem contributes to bridging the gap between unstructured textual information and structured financial analysis, enabling more comprehensive and informed decision-making in modern financial environments.

1.4. Research Objectives

1.4.1 Main Objective

The main objective of this research is to **design and develop a robust, domain-adapted sentiment analysis framework capable of accurately extracting, classifying, and quantifying sentiment from financial textual data in order to generate meaningful and actionable insights**.

This objective focuses on transforming unstructured financial text—such as news articles, reports, and disclosures—into structured sentiment indicators that reflect market perception and investor behavior. The framework is intended to address key challenges associated with financial sentiment analysis, including domain-specific language, contextual ambiguity, and variability in data sources.

Furthermore, the study aims to achieve a balance between **classification accuracy and model interpretability** by integrating multiple analytical approaches within a unified system. The objective also emphasizes the development of a scalable and adaptable solution that can remain effective across different financial contexts and evolving market conditions.

Ultimately, this research seeks to establish sentiment analysis as a reliable and practical tool for financial text mining, enabling improved understanding of qualitative information and supporting more informed decision-making processes in financial environments.

1.4.2 Specific Objectives

To achieve the main objective of this research, the following specific objectives are defined:

1. **To collect and construct a relevant financial textual dataset**
   Gather financial text data from credible sources such as news articles, corporate reports, and financial disclosures, ensuring adequate coverage of different market conditions and information types.

2. **To preprocess and standardize financial textual data**
   Apply Natural Language Processing (NLP) techniques including tokenization, stop-word removal, normalization, and text cleaning to prepare the data for analysis while preserving domain-specific meaning.

3. **To develop effective feature extraction mechanisms**
   Transform textual data into numerical representations using techniques such as Bag-of-Words and Term Frequency–Inverse Document Frequency (TF-IDF) to enable machine learning-based sentiment classification.

4. **To design and implement a hybrid sentiment classification model**
   Combine lexicon-based approaches with supervised machine learning algorithms to

classify financial text into sentiment categories (positive, negative, neutral), leveraging both interpretability and predictive performance.

5. **To evaluate the performance of the sentiment analysis model**
   Assess the effectiveness of the proposed framework using standard evaluation metrics such as accuracy, precision, recall, and F1-score, ensuring robustness and reliability.

6. **To analyze sentiment patterns and trends within financial text**
   Examine how sentiment varies across different data sources and time periods, identifying meaningful patterns that reflect market perception.

7. **To ensure model adaptability and scalability**
   Develop a framework that can handle diverse financial text sources and adapt to evolving language patterns and market dynamics.

8. **To provide a structured approach for transforming unstructured data into actionable insights**
   Demonstrate how extracted sentiment can be systematically interpreted and utilized to support financial analysis and decision-making processes.

These objectives collectively guide the development of a comprehensive sentiment analysis system tailored to financial textual data, ensuring both theoretical rigor and practical applicability.

2. METHODOLOGY

2.1 Understanding the key pillars of the research domain

The foundation of this research is built upon several core pillars that collectively define the domain of financial sentiment analysis. A clear understanding of these pillars is essential to ensure that the proposed methodology is both technically sound and aligned with the complexities of financial text mining. These pillars integrate concepts from Natural Language Processing (NLP), machine learning, and financial analytics, forming the basis for the overall system design.

The first key pillar is **Natural Language Processing (NLP)**, which provides the fundamental techniques required to process and analyze unstructured textual data. Financial text is inherently complex, containing domain-specific terminology, structured narratives, and contextual dependencies. NLP enables the transformation of raw text into a structured format through processes such as tokenization, normalization, and feature extraction. This pillar is critical as it directly influences the quality of input data used in sentiment classification.

The second pillar is **Sentiment Analysis**, which focuses on identifying and quantifying the emotional tone or polarity expressed within financial text. Unlike general sentiment analysis, financial sentiment requires careful interpretation due to the presence of neutral terminology, implicit meanings, and mixed sentiment expressions. This pillar defines how textual information is converted into measurable sentiment indicators, forming the core output of the system.

The third pillar is **Machine Learning and Classification Techniques**, which are used to model and predict sentiment based on extracted features. Supervised learning algorithms enable the system to learn patterns from labeled data and generalize to unseen text. The integration of machine learning enhances the system's ability to handle complex language structures and improves classification accuracy compared to rule-based approaches.

Another important pillar is **Domain Knowledge in Finance**, which ensures that the analysis is context-aware and relevant to financial applications. Financial text often includes specialized vocabulary, industry-specific concepts, and regulatory language that require domain understanding. Incorporating financial knowledge helps in refining preprocessing steps, improving sentiment interpretation, and reducing misclassification.

The fifth pillar is **Data Preprocessing and Feature Engineering**, which plays a crucial role in preparing textual data for analysis. High-quality preprocessing ensures that noise, inconsistencies, and irrelevant information are removed, while feature engineering determines how effectively textual information is represented for model training. This pillar directly impacts the performance and reliability of the sentiment analysis system.

Additionally, **Model Evaluation and Validation** forms a key pillar, ensuring that the developed system performs reliably and generalizes well to new data. Evaluation metrics such as accuracy, precision, recall, and F1-score are used to assess classification performance, while validation techniques help prevent overfitting and ensure robustness.

Finally, the pillar of **Practical Applicability and Scalability** ensures that the developed framework is not limited to theoretical analysis but can be deployed in real-world financial environments. This includes considerations such as handling large volumes of data, adapting to evolving market conditions, and maintaining computational efficiency.

In summary, the research domain is supported by interconnected pillars: NLP, sentiment analysis, machine learning, financial domain knowledge, data preprocessing, model evaluation, and system scalability. Understanding these pillars provides a structured foundation for designing an effective sentiment analysis framework tailored to financial textual data and guides the subsequent stages of the methodology.

2.2 Financial Text Sentiment Extraction and Classification Using a Hybrid Lexicon-Based and Machine Learning Framework

The data collection phase constitutes a critical component of the proposed sentiment analysis framework, as the quality, diversity, and relevance of the collected textual data directly influence the effectiveness of the overall system. In this study, financial textual data is systematically gathered from multiple credible and publicly available sources to ensure comprehensive coverage of market-related information.

The primary data sources include **financial news articles, corporate disclosures, earnings reports, and market analysis publications**. These sources are selected due to their significant influence on investor perception and their role in shaping market sentiment. News articles provide real-time insights into market events and external factors, while corporate disclosures

and financial reports offer structured and authoritative information regarding company performance and strategic direction.

Data collection is performed using automated techniques such as web scraping and API-based retrieval, where applicable. These methods enable efficient extraction of large volumes of textual data while maintaining consistency and scalability. During this process, relevant metadata such as publication date, source, and company identifiers are also captured to facilitate temporal analysis and data organization.

To ensure data quality, several filtering criteria are applied. Irrelevant content, duplicate entries, and incomplete records are removed to minimize noise and redundancy. Additionally, only English-language text is considered to maintain consistency in subsequent Natural Language Processing (NLP) tasks. The collected data is then stored in a structured format, enabling efficient access and processing in later stages of the methodology.

Another important aspect of this phase is the preparation of labeled data for supervised learning. Where labeled datasets are not readily available, manual or semi-automated annotation techniques are employed to assign sentiment labels (positive, negative, or neutral) to selected samples. This labeled subset is essential for training and evaluating machine learning models.

Overall, the data collection process is designed to ensure that the dataset is **representative, diverse, and high-quality**, capturing a wide range of financial sentiments across different contexts and time periods. This forms a strong foundation for the subsequent preprocessing, feature extraction, and sentiment classification stages of the research.

2.2.2 Methodology

This study adopts a **hybrid, multi-stage sentiment analysis framework** specifically designed for financial textual data. The methodology is inspired by advanced financial NLP architectures and incorporates both machine learning and lexicon-based techniques to address domain-specific challenges such as contextual ambiguity, class imbalance, and interpretability.

A. Problem Formulation and Labeling Strategy

Unlike traditional sentiment classification, which directly assigns polarity labels, this study reformulates sentiment detection as a **two-stage learning problem** to improve robustness and directional accuracy.

In financial contexts, not all textual information significantly impacts market behavior. Therefore, the first stage focuses on identifying whether a given text is **market-relevant (impactful)**, while the second stage determines the **direction of sentiment (positive, negative, neutral)**.

This approach is motivated by findings in prior architectures where single-stage classification resulted in poor performance due to class imbalance and dominance of neutral samples

B. Text Representation and Embedding

To capture semantic meaning beyond surface-level text, financial documents are transformed into **vector representations using embedding techniques**. These embeddings encode contextual relationships between words and enable the model to understand financial narratives more effectively.

The process includes:

- Converting preprocessed text into dense vector representations
- Capturing semantic similarity between financial events
- Enabling downstream classification tasks

This representation allows the model to move beyond simple keyword matching and instead understand **contextual sentiment**, which is critical in financial text.

C. Stage 1: Market Impact Detection Model

The first stage of the model is designed to determine whether a piece of financial text has a **significant impact on market behavior**.

A supervised machine learning model (e.g., gradient boosting such as XGBoost) is trained to estimate:

- Whether a news item or document is **market-moving** or not

This stage acts as a **filtering mechanism**, ensuring that only relevant financial text proceeds to sentiment classification.

To improve sensitivity:

- A **recall-oriented threshold** is applied
- The model prioritizes detecting impactful events, even at the cost of some false positives

This design choice is crucial because missing impactful information is more costly than misclassifying non-impactful text.

D. Stage 2: Sentiment Direction Classification

For text identified as impactful, the second stage performs **sentiment classification**, predicting whether the sentiment is:

- Positive
- Negative
- Neutral

This stage uses supervised learning models trained on labeled financial text data. However, as identified in prior work, directional prediction is inherently more difficult than impact detection, with models often showing near-random performance when used alone .

To address this limitation, additional enhancements are introduced.

E. Lexicon-Based Sentiment Enhancement

To improve interpretability and robustness, a **domain-specific financial lexicon** is integrated into the model.

The lexicon assigns sentiment scores to financial terms, and a document-level score is computed as:

- Aggregation of individual word sentiment values
- Adjustment for financial context and polarity

This component is particularly useful when:

- Model confidence is low
- Text contains explicit sentiment indicators

As highlighted in the reference architecture, lexicon-based scoring is used as a **fallback or complementary mechanism** when model predictions are uncertain

F. Hybrid Decision Mechanism

A key innovation in the methodology is the **adaptive combination of machine learning predictions and lexicon-based scores**.

Instead of relying solely on one method, the final sentiment prediction is obtained through:

- Model probability outputs
- Lexicon-derived sentiment confidence
- A weighted fusion mechanism

This hybrid approach:

- Improves classification accuracy
- Maintains interpretability
- Reduces dependency on large labeled datasets

G. Handling Class Imbalance

Financial sentiment datasets typically exhibit **class imbalance**, where neutral sentiment dominates.

To mitigate this issue:

- The problem is decomposed into two stages (impact + direction)
- Threshold tuning is applied
- Model evaluation focuses on **F1-score and recall**, rather than accuracy alone

This restructuring significantly improves the model's ability to detect minority classes (positive and negative sentiment).

H. Semantic Similarity and Context Retrieval (Optional Enhancement)

To enhance interpretability and contextual understanding, the methodology incorporates a **semantic similarity mechanism**:

- Text embeddings are compared using similarity measures (e.g., cosine similarity)
- Historical documents with similar semantic meaning are retrieved

As described in the reference system, this enables:

- Identification of past events with similar sentiment patterns
- Contextual explanation of model predictions

This component adds an explainability layer by linking predictions to real historical examples.

I. Model Evaluation

The performance of the sentiment model is evaluated using multiple metrics:

- **Accuracy** – overall correctness
- **Precision** – correctness of positive predictions
- **Recall** – ability to detect relevant sentiment
- **F1-score** – balance between precision and recall
- **ROC-AUC** – classification capability across thresholds

Special emphasis is placed on:

- Performance on minority classes
- Robustness of directional predictions

J. Summary of the Sentiment Model Pipeline

The complete sentiment analysis workflow is summarized as:

Text Input → Preprocessing → Embedding → Stage 1 (Impact Detection) → Stage 2 (Sentiment Classification) → Lexicon Enhancement → Hybrid Decision → Output Sentiment

---

K. Key Contributions of the Methodology

The proposed sentiment model introduces several improvements over traditional approaches:

- A **two-stage architecture** to handle imbalance and improve robustness
- A **hybrid model** combining machine learning and lexicon methods
- Enhanced **interpretability through lexicon scoring and similarity retrieval**
- Adaptation to **financial domain-specific language and context**

2.4 High-Level Architecture Diagram

2.5 Project Requirements that have been achieved

2.5.1 Functional Requirements

1. Financial Text Data Ingestion
The system is capable of collecting and ingesting financial textual data from multiple sources such as news articles, reports, and disclosures. It supports automated data retrieval and ensures that relevant textual information is available for analysis.

---

2. Text Preprocessing and Cleaning
The system performs comprehensive preprocessing of raw textual data, including noise removal, tokenization, normalization, and stop-word elimination. This ensures that the input data is standardized and suitable for further processing.

---

3. Feature Extraction and Representation
The system converts processed text into numerical representations using techniques such as TF-IDF and n-grams. This enables effective input for machine learning models and improves classification performance.

---

4. Market Impact Detection
The system implements a first-stage classification model to identify whether a given piece of financial text has a significant impact on market behavior. This filtering mechanism ensures that only relevant information is considered for sentiment analysis.

---

5. Sentiment Classification
The system classifies financial text into sentiment categories (positive, negative, and neutral)

using supervised machine learning models. This provides structured sentiment outputs from unstructured text data.

---

6. Lexicon-Based Sentiment Scoring
A domain-specific financial lexicon is integrated into the system to compute sentiment scores based on predefined word polarity. This enhances interpretability and supports model predictions.

---

7. Hybrid Sentiment Decision Mechanism
The system combines machine learning predictions with lexicon-based scores to produce final sentiment outputs. This hybrid approach improves robustness and accuracy.

---

8. Handling Class Imbalance
The system incorporates a two-stage classification approach and threshold tuning to address class imbalance issues commonly found in financial datasets.

---

9. Model Evaluation and Validation
The system evaluates model performance using metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. This ensures reliability and effectiveness of the sentiment classification process.

---

10. Sentiment Output Generation
The system produces structured sentiment outputs that can be used for analysis, visualization, or integration into decision-making processes.

2.5.2 Non-Functional Requirements

In addition to functional capabilities, the developed sentiment analysis system satisfies several non-functional requirements that ensure the system operates efficiently, reliably, and is suitable for practical deployment. These requirements define the quality attributes of the system and support its overall performance and usability.

---

1. Performance Efficiency
The system is designed to process large volumes of financial textual data within a reasonable

time frame. Efficient preprocessing pipelines and optimized machine learning models ensure low latency in sentiment prediction, making the system suitable for near real-time analysis.

---

2. Scalability
The architecture supports scalability to handle increasing data volumes and additional data sources. The modular design allows components such as data ingestion, preprocessing, and classification to be scaled independently based on workload requirements.

---

3. Accuracy and Reliability
The system maintains high classification accuracy through the use of a hybrid model combining machine learning and lexicon-based approaches. Robust evaluation techniques and validation methods ensure consistent and reliable performance across different datasets.

---

4. Interpretability and Explainability
Given the importance of transparency in financial applications, the system provides interpretable outputs. Lexicon-based scoring and model insights enable users to understand how sentiment decisions are derived, supporting trust and accountability.

---

5. Maintainability
The system is structured using a modular design, allowing easy updates, debugging, and extension of individual components. This ensures that improvements or modifications can be implemented without affecting the entire system.

---

6. Flexibility and Adaptability
The system is adaptable to different types of financial text sources and evolving language patterns. New data sources, updated lexicons, or improved models can be integrated with minimal restructuring.

---

7. Robustness
The system is capable of handling noisy, incomplete, or inconsistent textual data. Preprocessing techniques and validation mechanisms ensure stable performance even under imperfect data conditions.

---

8. Usability
The system produces clear and structured outputs that are easy to interpret and utilize. This ensures that users, including analysts and researchers, can effectively leverage sentiment insights without requiring deep technical expertise.

---

9. Resource Efficiency
The methodology balances computational complexity and performance by combining lightweight lexicon methods with more advanced machine learning models. This ensures efficient use of computational resources.

---

10. Reproducibility
The system ensures that results can be consistently reproduced through well-defined preprocessing steps, model configurations, and evaluation procedures. This is essential for research validity and future improvements.

---

11. Security and Data Integrity
Collected data is handled in a controlled manner, ensuring that data integrity is maintained throughout the processing pipeline. Proper data handling mechanisms prevent corruption or loss of information.

---

12. Extensibility
The system is designed to support future enhancements such as integration with deep learning models, real-time data streams, or additional financial analytics modules.

In summary, the non-functional requirements achieved by the system ensure that it is not only technically effective but also practical, scalable, and reliable for real-world financial sentiment analysis applications.

2.5.3 Other Requirements

In addition to the functional and non-functional requirements, several supporting requirements were identified and achieved to ensure the successful development, deployment, and usability of the sentiment analysis system within a research and practical context.

---

1. Data Availability and Accessibility
The system requires access to reliable and up-to-date financial textual data sources. Publicly available datasets and financial news platforms are utilized to ensure continuous data availability. The system is designed to accommodate different data formats and sources without significant restructuring.

---

2. Development Environment Requirements
The implementation is carried out using standard development tools and programming environments suitable for machine learning and NLP tasks. This includes support for libraries and frameworks required for text processing, model training, and evaluation.

---

3. Hardware Requirements
The system is designed to operate on standard computing hardware without requiring specialized high-performance infrastructure. However, for larger datasets or advanced models, higher computational resources (e.g., increased RAM or GPU support) may enhance performance.

---

4. Software Dependencies
The system depends on widely used software libraries for Natural Language Processing and machine learning. These dependencies are selected based on stability, community support, and compatibility to ensure long-term usability and maintainability.

---

5. Data Storage and Management
Efficient storage mechanisms are implemented to manage collected textual data and processed outputs. Structured storage formats are used to enable easy retrieval, updating, and analysis of data.

---

6. Integration Capability
The system is designed to allow integration with other analytical modules or external systems. This enables the sentiment outputs to be used in broader applications such as financial forecasting, dashboards, or decision-support systems.

---

7. Documentation and Reproducibility
Comprehensive documentation is maintained for all stages of the system, including data

collection, preprocessing, model development, and evaluation. This ensures that the research can be reproduced and extended in future work.

---

8. Ethical Considerations
The system uses publicly available data and adheres to ethical standards in data usage. Care is taken to avoid misuse of information and ensure that the analysis does not violate any data privacy or usage policies.

---

9. Testing and Validation Requirements
The system undergoes systematic testing to verify correctness, stability, and performance. Both unit-level and system-level validations are conducted to ensure that each component functions as expected.

---

10. Deployment Considerations
Although primarily developed for research purposes, the system is designed with potential deployment in mind. This includes considerations for scalability, modular integration, and adaptability to real-world financial environments.

---

11. Version Control and Collaboration
Version control mechanisms are utilized to manage code changes and support collaborative development. This ensures traceability, consistency, and efficient teamwork throughout the project lifecycle.

---

12. Future Enhancement Support
The system is designed to support future improvements, including the incorporation of deep learning models, real-time data processing, and advanced analytics. This ensures that the system remains relevant and extendable beyond the current scope.

2.6. Consideration of the aspects of the system

2.6.1 Social aspects

The development and deployment of a financial sentiment analysis system carry several social implications, particularly in how information is interpreted, disseminated, and utilized by individuals and institutions. These aspects are important to ensure that the system contributes positively to society while minimizing potential risks.

One key social aspect is the **democratization of financial information**. Traditionally, access to advanced analytical tools and market insights has been limited to institutional investors with significant resources. The proposed sentiment analysis system helps bridge this gap by transforming complex financial text into understandable sentiment indicators. This enables retail investors and smaller stakeholders to access insights that were previously difficult to obtain, thereby promoting more inclusive participation in financial markets.

Another important consideration is the **influence on investor behavior**. Sentiment analysis systems can shape decision-making by highlighting positive or negative market signals. While this can support more informed decisions, there is also a risk that users may over-rely on automated sentiment outputs without critical evaluation. This could lead to herd behavior or amplified market reactions. Therefore, the system is designed to provide **supportive insights rather than prescriptive decisions**, encouraging users to combine sentiment analysis with other forms of financial evaluation.

The issue of **information credibility and bias** is also significant. Financial text data may contain biased, incomplete, or misleading information depending on the source. If not properly managed, these biases can propagate through the sentiment analysis model and influence outcomes. To mitigate this, the system incorporates data filtering and preprocessing mechanisms, and emphasizes the use of credible sources. However, users must remain aware that sentiment outputs reflect the input data and are not absolute truths.

Transparency and interpretability play a crucial role in building user trust. In financial contexts, users need to understand how conclusions are derived. The integration of lexicon-based scoring alongside machine learning models enhances explainability, allowing users to see the contributing factors behind sentiment classifications. This reduces the perception of the system as a “black box” and promotes responsible usage.

Another social consideration is the **impact on employment and professional roles**. The adoption of automated sentiment analysis tools may reduce reliance on manual analysis performed by financial analysts. However, rather than replacing human expertise, the system is intended to augment analytical capabilities by handling large volumes of data efficiently, allowing professionals to focus on higher-level decision-making.

The system also considers **ethical data usage**. Only publicly available financial text is utilized, and no personal or sensitive data is processed. This ensures compliance with ethical standards and avoids privacy concerns. Additionally, the system does not manipulate or generate misleading information but strictly analyzes existing data.

Finally, there is the aspect of **financial literacy and accessibility**. By presenting sentiment insights in a structured and understandable manner, the system can contribute to improving users’ understanding of market dynamics. However, care must be taken to ensure that outputs are not oversimplified in a way that misleads users about the complexity of financial markets.

In summary, the social aspects of the system highlight its potential to enhance accessibility, support informed decision-making, and improve financial literacy, while also acknowledging

risks related to bias, over-reliance, and information interpretation. Addressing these considerations ensures that the system is used responsibly and contributes positively to the broader financial ecosystem.

Security is a critical consideration in the design and implementation of the sentiment analysis system, particularly due to its reliance on data collection, processing, and storage. Although the system primarily operates on publicly available financial textual data, ensuring the integrity, confidentiality, and reliability of the system remains essential.

One of the primary security concerns is **data integrity**. The system must ensure that the collected financial text data is not altered, corrupted, or tampered with during collection, storage, or processing. To address this, validation mechanisms are applied during data ingestion, including filtering malformed inputs and verifying data consistency. This ensures that sentiment analysis results are based on accurate and trustworthy data.

Another important aspect is **secure data handling and storage**. The system organizes and stores textual data and processed outputs in a structured manner to prevent data loss or unauthorized modification. Access to stored data is controlled to ensure that only authorized processes or users can retrieve or modify information. Proper backup mechanisms are also considered to safeguard against accidental data loss.

The system also addresses **protection against malicious input and attacks**. Since data may be collected from external sources, there is a risk of encountering harmful or manipulated content. Input validation and preprocessing steps are designed to detect and filter out potentially harmful data, such as scripts or malformed text, thereby reducing the risk of system vulnerabilities such as injection attacks.

System reliability and availability are also closely tied to security. The system is designed to operate consistently without interruptions, even when processing large volumes of data. Mechanisms such as error handling and fault tolerance are implemented to ensure that failures in one component do not compromise the entire system.

Another key consideration is **model security and robustness**. Machine learning models can be vulnerable to adversarial inputs or biased data that may affect predictions. While the system primarily operates in a research context, care is taken to ensure that the model is trained on clean and representative data, reducing the risk of manipulation or misleading outputs.

Access control and authorization are also important, especially if the system is extended to a deployed environment. Role-based access mechanisms can be implemented to restrict system functionalities based on user roles, ensuring that sensitive operations such as model updates or data modification are controlled.

In addition, **secure integration and communication** are considered when the system interacts with external data sources or other systems. Ensuring safe data transfer mechanisms helps prevent unauthorized interception or data leakage.

Finally, the system adheres to **ethical and responsible data usage practices**, ensuring that only publicly available and non-sensitive financial data is used. This minimizes privacy risks and aligns the system with acceptable data governance standards.

In summary, the security aspects of the system focus on maintaining data integrity, protecting against malicious inputs, ensuring reliable operation, and enabling controlled access. These measures collectively contribute to building a secure and trustworthy sentiment analysis system suitable for financial applications.

2.6.3 Ethical aspects

The development and deployment of the financial sentiment analysis system involve several ethical considerations to ensure that the system is used responsibly, transparently, and without causing harm to users or the broader financial ecosystem. These considerations are particularly important given the potential influence of sentiment-based insights on investment decisions and market behavior.

One of the primary ethical aspects is **responsible use of data**. The system relies exclusively on publicly available financial textual data such as news articles and corporate disclosures. No personal, private, or sensitive data is collected or processed. This ensures compliance with ethical data usage standards and avoids privacy violations. Additionally, proper attribution of data sources is maintained where applicable.

Another important consideration is **bias and fairness**. Financial text data may contain inherent biases depending on the source, publication style, or market perspective. If not properly addressed, these biases can propagate through the sentiment analysis model and lead to skewed or misleading results. To mitigate this, the system incorporates preprocessing and filtering mechanisms and uses diverse data sources where possible. However, users are informed that the outputs reflect the underlying data and may not represent an objective truth.

The issue of **transparency and explainability** is also central to ethical system design. In financial applications, users must understand how conclusions are derived. The system addresses this by integrating interpretable components such as lexicon-based sentiment scoring alongside machine learning models. This allows users to trace the reasoning behind sentiment classifications, reducing reliance on opaque “black-box” decisions.

Another key ethical concern is **misuse and over-reliance**. There is a risk that users may treat sentiment outputs as definitive investment advice rather than analytical support. This could lead to poor financial decisions or contribute to herd behavior in markets. To address this, the system is designed as a **decision-support tool**, not a decision-making authority. Users are expected to combine sentiment insights with other forms of financial analysis and professional judgment.

The system also considers the **impact on market dynamics**. Large-scale use of sentiment analysis tools could amplify reactions to news and potentially contribute to market volatility. While the system itself does not directly influence markets, its outputs could indirectly affect

user behavior. Therefore, it is important that the system is used responsibly and that its limitations are clearly communicated.

Accountability is another important aspect. The system provides analytical outputs based on data and models, but it does not assume responsibility for decisions made by users based on those outputs. Clear documentation and explanation of system limitations are provided to ensure that users understand the scope and boundaries of the system.

Additionally, the system adheres to **academic integrity and research ethics**. All methodologies, data handling procedures, and results are documented transparently to ensure reproducibility and credibility. Any external sources or references are properly acknowledged.

Finally, the system promotes **ethical innovation**, aiming to enhance financial analysis without replacing human judgment or encouraging irresponsible behavior. It is designed to augment human decision-making rather than automate it entirely.

In summary, the ethical aspects of the system focus on responsible data usage, bias mitigation, transparency, prevention of misuse, and accountability. Addressing these considerations ensures that the sentiment analysis system is aligned with ethical standards and contributes positively to both research and practical financial applications.

Forecasting Dividend Behavior of Firms Listed on the Colombo Stock Exchange

Project Id: 25-26J-445

Project Final Report

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DECLARATION

I certify that this is my own work, and that this thesis does not incorporate without acknowledgment any material previously submitted for a degree or diploma at any other university or Institute of higher learning, and that it does not includes any material previously released or written by another person, except where acknowledgement is made in the text.

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ABSTRACT

Dividend forecasting plays a crucial role in investment decision-making, particularly in emerging markets such as Sri Lanka, where investors rely heavily on dividend income for portfolio stability and long-term returns; however, predicting dividend behavior in the Colombo Stock Exchange (CSE) is complex due to the combined influence of firm-level financial indicators and macroeconomic factors. Traditional forecasting approaches based on statistical and econometric models are limited in capturing non-linear relationships and often lack practical applicability for investors. This research proposes the development of an AI-driven Dividend Forecasting and Analysis System designed as a decision-support tool to analyze and predict dividend behavior of CSE-listed companies. The system integrates historical dividend data, company financial indicators, and macroeconomic variables to identify patterns and forecast trends such as whether a firm's dividend payout is likely to increase, decrease, or remain stable in the upcoming period. Additionally, the system introduces scenario-based analysis, enabling users to adjust macroeconomic factors such as inflation, interest rates, and GDP growth to evaluate their potential impact on dividend decisions. The proposed solution will be implemented as an interactive web-based dashboard, allowing investors, analysts, and policymakers to explore historical trends, compare companies, and make informed, data-driven investment decisions. By combining machine learning techniques with a user-oriented analytical interface, this research aims to bridge the gap between theoretical financial models and practical decision-making, ultimately enhancing the ability to anticipate dividend behavior within the Sri Lankan capital market.

Keywords: Dividend Forecasting, Machine Learning, Colombo Stock Exchange (CSE), Decision Support System, Financial Analytics, Emerging Markets

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LIST OF ABBREVIATIONS

CSE	Colombo Stock Exchange
ML	Machine Learning
AI	Artificial Intelligence
ML	Machine Learning
RF	Random Forest
XGBoost	Extreme Gradient Boosting
LR	Logistic Regression
GDP	Gross Domestic Product
EPS	Earnings Per Share
ROE	Return on Equity
SDLC	Software Development Life Cycle
WBS	Work Breakdown Structure

INTRODUCTION

General Introduction

Dividends represent a key component of investment returns and serve as an important indicator of a company's financial health and stability. In emerging markets such as Sri Lanka, investors often rely on dividend income as a primary source of return, especially in environments characterized by market volatility and economic uncertainty. Within the Colombo Stock Exchange (CSE), dividend-paying companies are generally perceived as more reliable and attractive, as consistent dividend payments signal profitability, effective financial management, and long-term sustainability.

However, understanding and predicting dividend behavior is a complex task. Dividend decisions are influenced by a combination of internal financial factors such as earnings, liquidity, leverage, and growth opportunities, as well as external macroeconomic conditions including inflation, interest rates, and overall economic performance. These interconnected factors make dividend behavior dynamic and difficult to analyze using simple or traditional approaches. As a result, investors often face challenges in anticipating whether a company's dividend payout will increase, decrease, or remain stable in future periods.

In practice, most investors in the Sri Lankan market depend on manual analysis of financial statements, historical trends, and external reports to make dividend-related decisions. This process is time-consuming, requires significant financial knowledge, and may still lead to inaccurate conclusions due to the complexity of the underlying relationships. Furthermore, existing research in the Sri Lankan context has primarily focused on identifying factors that influence dividend policies rather than developing practical tools that support real-world decision-making.

With advancements in data analytics and machine learning, there is an opportunity to move beyond traditional analysis toward predictive and data-driven solutions. Machine learning techniques are capable of capturing complex, non-linear relationships between financial and macroeconomic variables, making them suitable for analyzing dividend behavior in dynamic market conditions. Leveraging these capabilities, this research proposes the development of an AI-driven decision-support system that assists investors in understanding and forecasting dividend behavior of CSE-listed companies.

The proposed system integrates historical dividend data, company financial indicators, and macroeconomic variables to provide insights into future dividend trends. Through an interactive web-based dashboard, users will be able to explore patterns, analyze influencing factors, and evaluate how changes in economic conditions may impact dividend decisions. By focusing on practical usability

and real-world application, this research aims to enhance investor decision-making and contribute to a more data-driven investment environment within the Sri Lankan capital market.

Background literature

Dividend policy has long been a central topic in corporate finance, as it reflects how companies distribute earnings to shareholders while balancing reinvestment needs. Several classical theories provide a foundation for understanding dividend behavior. The Dividend Irrelevance Theory suggests that dividend decisions do not affect firm value under perfect market conditions, while the Signaling Theory proposes that dividends convey information about a firm's future prospects. Agency Theory highlights dividends as a mechanism to reduce conflicts between managers and shareholders, and the Bird-in-the-Hand Theory argues that investors prefer certain dividend income over uncertain capital gains.

Although these theories offer valuable insights, their assumptions often do not hold in real-world contexts, particularly in emerging markets such as Sri Lanka. In such environments, dividend decisions are influenced not only by firm-level financial performance but also by broader economic conditions, regulatory frameworks, and market uncertainties. Factors such as profitability, liquidity, leverage, and firm size, along with macroeconomic variables like inflation, interest rates, and economic growth, collectively shape dividend behavior. As a result, dividend policy in the Colombo Stock Exchange (CSE) reflects a complex interaction of internal and external influences rather than a single theoretical perspective.

Despite the importance of dividends in investment decision-making, existing approaches in Sri Lanka largely focus on descriptive and explanatory analysis. These approaches identify relationships between variables but provide limited support for anticipating future dividend behavior. This creates a need for more advanced and practical methods that can assist investors in understanding how dividend patterns may evolve over time.

A considerable body of research has examined dividend policy and its determinants both in Sri Lanka and globally. In the Sri Lankan context, empirical studies have identified key factors influencing dividend decisions, including profitability, earnings stability, liquidity, and ownership structure. These studies primarily rely on econometric models such as regression analysis to explain how these variables affect dividend payouts. While such findings contribute to theoretical understanding, they

remain limited in their ability to provide forward-looking insights or support real-time investment decisions.

In contrast, recent global research has explored the application of machine learning techniques in financial forecasting, including dividend behavior analysis. Machine learning models such as Random Forest, gradient boosting algorithms, and neural networks have demonstrated improved performance over traditional statistical methods due to their ability to capture complex and non-linear relationships among variables. Studies conducted in various emerging markets have shown that these models can effectively classify dividend-related outcomes and identify patterns that are not easily detected through conventional approaches.

However, despite these advancements, several limitations remain. Many machine learning-based studies focus primarily on classification tasks, such as predicting whether a company will pay dividends, without extending their findings into practical tools for investors. Furthermore, most existing research does not adequately incorporate macroeconomic scenario analysis, which is particularly important in volatile economic environments like Sri Lanka. The absence of user-oriented systems means that these models are often confined to academic settings and are not accessible for real-world decision-making.

Additionally, there is a noticeable lack of research specifically tailored to the Colombo Stock Exchange that integrates financial indicators, macroeconomic variables, and predictive analytics into a unified framework. Existing solutions do not provide interactive capabilities that allow users to explore how changes in economic conditions may influence dividend behavior.

Therefore, while prior studies offer valuable theoretical and methodological contributions, there remains a clear gap in translating these insights into practical, decision-support systems. This research aims to address this gap by developing an AI-driven platform that not only analyzes dividend behavior but also enables users to make informed predictions about future dividend trends in a dynamic and interactive manner.

Research Gap

Despite the extensive body of research on dividend policy, several important limitations can be identified, particularly within the context of the Colombo Stock Exchange (CSE). Existing studies in Sri Lanka primarily focus on identifying the determinants of dividend decisions using traditional econometric approaches. While these studies provide valuable insights into the factors influencing dividend behavior, they are largely descriptive in nature and do not offer predictive capabilities that can support forward-looking investment decisions.

Furthermore, although global research has increasingly adopted machine learning techniques for financial forecasting, these approaches are often limited to experimental or academic settings. Many of these models focus on isolated prediction tasks without being integrated into practical, user-oriented systems. As a result, there is a lack of accessible tools that translate analytical findings into actionable insights for investors.

Another significant limitation is the insufficient incorporation of macroeconomic variables in predictive analysis. Dividend behavior in emerging markets is highly sensitive to economic conditions such as inflation, interest rates, and economic growth; however, many existing models do not provide mechanisms to evaluate how changes in these factors may influence future dividend trends.

Additionally, there is no comprehensive system specifically designed for the Sri Lankan market that combines historical dividend data, firm-level financial indicators, and macroeconomic variables into a unified analytical framework. The absence of interactive platforms further limits the ability of investors to explore patterns, compare companies, and assess potential future outcomes in a structured manner.

Therefore, a clear research gap exists in the development of a practical, data-driven decision-support system that enables users to analyze and anticipate dividend behavior within the Colombo Stock Exchange.

RESEARCH PROBLEM

Dividend behavior plays a critical role in investment decision-making, yet accurately anticipating changes in dividend patterns remains a significant challenge in the Colombo Stock Exchange (CSE). Investors often depend on manual analysis of financial statements, historical trends, and external reports to assess whether a company's dividend payout is likely to increase, decrease, or remain stable. This process is not only time-consuming but also requires a high level of financial expertise, and it may still lead to inaccurate conclusions due to the complexity and dynamic nature of influencing factors.

Existing approaches in Sri Lanka are predominantly based on traditional statistical methods, which are limited in their ability to capture non-linear relationships between financial and macroeconomic variables. As a result, these methods are not well-suited for predicting dividend behavior in a rapidly changing economic environment. Moreover, while machine learning techniques have shown potential in improving prediction accuracy, their application in the Sri Lankan context remains limited, and they are rarely implemented as practical tools for end users.

Another key issue is the lack of integration between financial data and macroeconomic conditions when analyzing dividend trends. Investors currently do not have access to systems that allow them to evaluate how changes in economic factors, such as inflation or interest rates, may impact future dividend decisions. This lack of comprehensive and interactive analysis further complicates the decision-making process.

Consequently, there is a need for a system that can bridge the gap between complex financial data and practical investment decision-making. Specifically, a solution is required to support investors in understanding and predicting dividend behavior in a structured, data-driven manner. This research therefore addresses the problem of how to develop an effective decision-support system that leverages machine learning techniques to analyze historical patterns, incorporate macroeconomic factors, and provide meaningful insights into the future dividend behavior of CSE-listed companies.

COMPREHENSIVE FINANCIAL HEALTH ANALYSIS MODULE FOR COLOMBO STOCK EXCHANGE (CSE) LISTED COMPANIES

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DECLARATION OF THE CANDIDATE AND SUPERVISOR

I declare that this is my own work and this dissertation does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the dissertation with the following declaration.

The above candidate has carried out research for the bachelor's degree Dissertation under my supervision.

Signature of the Supervisor: ……..

Date: ……

Signature of the Co-supervisor: …….

Date: ……

ABSTRACT

Colombo Stock Exchange (CSE) is an institution that forms the backbone of the Sri Lankan capital market; however, investors find it difficult to determine the level of financial sustainability of listed companies because of inaccessible, opaque, and unreliable valuation tools locally. The current platforms were originally created and optimised to serve global markets, specifically NYSE and NASDAQ, and are therefore inappropriate to the economic conditions of Sri Lanka, its reporting regulations, and disclosure practices.

This study is aimed at developing a Comprehensive Financial Health Analysis Module uniquely tailored to CSE-listed companies. The proposed system will ease the extraction and preprocessing of financial disclosures, avoiding data heterogeneity and incompleteness across balance sheets, income statements, and cash flow statements. It will then calculate key financial ratios to provide a standardised basis for evaluating company performance levels. With the intent of delivering forward-looking insights, the module will utilise machine learning models — specifically Support Vector Regression (SVR) — combined with Explainable AI (XAI) techniques using SHAP (SHapley Additive Explanations), which will empower investors not only to detect financial risks but also to comprehend the causal factors behind those risks.

The solution will be delivered via an interactive web-based analysis platform providing both trend statistics and on-going analysis. The proposed project will bridge the gap between legally mandated financial disclosures and actionable investment intelligence for Sri Lankan investors, empowering them with accurate, context-sensitive, and credible financial insights that will ultimately yield informed decision-making and enhanced confidence in the capital market.

Keywords: *Colombo Stock Exchange, Financial Health Analysis, Machine Learning, Support Vector Regression, Explainable AI, SHAP, Z-Score, Financial Risk Prediction*

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
API	Application Programming Interface
AWS	Amazon Web Services
CSE	Colombo Stock Exchange
GCP	Google Cloud Platform
MAE	Mean Absolute Error
ML	Machine Learning
NLP	Natural Language Processing
RMSE	Root Mean Square Error
ROA	Return on Assets
ROE	Return on Equity
SHAP	SHapley Additive ExPlanations
SLIIT	Sri Lanka Institute of Information Technology
SVR	Support Vector Regression
WBS	Work Breakdown Structure
XAI	Explainable Artificial Intelligence
XBRL	eXtensible Business Reporting Language

1. INTRODUCTION

Investment decision-making in the Colombo Stock Exchange (CSE) is highly affected by the lack of access to complex financial analysis instruments as well as the specific economic dynamics of a developing economy. Although investors in developed markets have had access to sophisticated tools such as Bloomberg Terminal and Reuters Eikon, investors in Sri Lanka — both retail and institutional — do not have access to comprehensive and locally relevant investment assessment systems capable of evaluating the financial health of listed companies and the degree of investment risk.

Conventional financial analysis tools place much weight on manual analysis of financial reports, a process that is tedious, susceptible to human error, and too cumbersome to apply across the 300+ firms listed on the CSE. Furthermore, current global financial analysis platforms do not adequately capture the set of factors that define the financial environment in Sri Lanka in terms of regulation, economic indicators, and market trends. This leaves a significant shortage of investment intelligence customised to local market participants.

The emergence of novel breakthroughs in financial technology has indicated the prospect of machine learning to revolutionise investment analysis. Such technologies allow large volumes of financial data to be processed automatically, complex risk modelling to be executed, and understandable insights to be created to support investment decision-making. Nonetheless, the utilisation of these novel methodologies within the context of the CSE has been relatively untapped and represents a major avenue of innovation for the Sri Lankan financial services sector.

This research project aims to address these challenges by developing a Comprehensive Financial Health Analysis Module specifically designed for CSE-listed companies. The system will combine automated data acquisition from official

CSE sources, comprehensive computation of financial ratios, and machine learning–based risk prediction models using Support Vector Regression (SVR). Explainable AI (XAI) techniques employing SHAP values will ensure that risk predictions are transparent and interpretable.

The proposed system will fill the gap between advanced financial analysis systems available in developed countries and the particular requirements of Sri Lankan investors. This study will add value to investment decision-making, market disclosure, and ultimately effective capital allocation in the Sri Lankan economy by offering localised, explainable, and comprehensive financial health measures.

This project constitutes a complete individual effort entailing the design, development, and evaluation of the entire financial health analysis pipeline. The work involves obtaining and cleaning financial data for CSE-listed companies, computing the necessary financial risk ratios across liquidity, leverage, profitability, and efficiency dimensions, and applying regression-based machine learning models to predict continuous financial risk scores (Z-score), enabling more stable and accurate forecasting of financial health.

1.1 Background and Literature Survey

Financial health evaluation of stock markets has tremendously transformed from conventional manual evaluation systems to advanced automated systems utilising machine learning and artificial intelligence. Within the framework of an emerging market — specifically, the Colombo Stock Exchange (CSE) — this development brings both the prospects of enhanced analytical capability and the issues peculiar to developing financial environments.

The conventional method of financial health evaluation has been based on manual analysis of financial reports comprising balance sheets, income statements, and cash flow statements, along with the calculation of major financial ratios covering

liquidity, leverage, profitability, and efficiency dimensions [1]. Such a traditional approach is rife with disadvantages including time-consuming procedures, difficulties in cross-company comparison, and the inability to ensure analytical accuracy when large numbers of listed companies are involved [2].

Modern studies reveal that emerging markets have different peculiarities that necessitate special analytical models distinct from those developed for mature markets [3]. Research conducted to determine how to predict financial distress in third-world economies indicates that leverage, profitability, economic growth, company size, and interest rates are all important factors influencing the probability of companies transitioning to various financial health statuses [2]. These results underscore the need for localised analytical tools that account for the specific regulatory, economic, and operational environment peculiar to emerging markets. South Asian market-specific research shows that operational risk and liquidity risk have statistically significant positive effects on financial performance, whereas credit risk exhibits negative effects, and these complex risk dynamics need to be captured in full-scale financial health assessment schemes.

The introduction of machine learning algorithms to financial analysis has emerged as a ground-breaking method to overcome the shortcomings of traditional approaches. Recent studies highlight that regression-based models such as Support Vector Regression (SVR) provide more stable and reliable predictions for continuous financial risk forecasting, especially in emerging markets where non-linearity and data limitations are prevalent [3][4][5]. These algorithms are especially useful in such contexts because they help capture non-linear financial relationships that cannot be identified using traditional linear models.

Financial data preprocessing and standardisation has become an essential aspect of building sound automated analysis systems. Research on automated financial statement processing has shown that sophisticated systems can reduce manual

processing requirements by up to 80 percent while significantly enhancing data quality and reporting speed [6]. Studies underline the importance of clean, valid, and standardised data in making cross-company comparisons meaningful, particularly in markets with diverse reporting habits and styles [7]. Well-designed data preprocessing pipelines can significantly decrease error rates and improve the overall quality of analysis.

The research on financial distress prediction models has been widely explored in the academic literature, with both traditional methods — including the Altman Z-Score and their evolution into machine learning methods. Systematic reviews differentiate the Altman Z-Score as having an accuracy of approximately 90.91% in conventional models [8], making it one of the most widely used tools in both manufacturing and banking sectors. However, comparative studies have identified that machine learning-based solutions significantly outperform traditional ones, with hybrid frameworks combining conventional financial metrics and advanced algorithms achieving prediction accuracy exceeding 99 percent [9]. Research on XGBoost-based predictive financial risk models shows accuracy rates of 84.77% when predicting systemic financial risks, while broader machine learning model studies report success rates exceeding 96% in correctly identifying financially troubled companies [10].

Research examining the correlation between financial indicators and market performance in emerging economies confirms that earnings per share and net asset value per share exhibit strong correlations with market valuation, justifying the significance of comprehensive financial statement analysis [11]. Analysis of plantation companies listed on emerging market exchanges demonstrates a strong correlation between financial distress indicators and performance results, with financially healthy companies consistently exhibiting superior performance outcomes [12]. These results justify the creation of complex analytical models that reflect the intricate relationships between financial measurements and market performance in emerging market contexts.

A standardisation of financial reporting in emerging markets has been cited as a key element facilitating automated analysis. Studies on financial statement comparability indicate that firms with higher comparability enjoy lower costs of equity, implying significant economic gains from standardised reporting practices [13]. The adoption of digital reporting standards, including the CSE's implementation of XBRL in 2025, addresses many data standardisation issues and enables more effective automated analysis systems.

Systematic studies of machine learning implementation in financial health assessment show that ensemble techniques and hybrid methods are superior to single-algorithm implementations. Comparative studies confirm that Random Forest and Gradient Boosting models perform most efficiently across diverse financial predictive tasks [14], while research investigating multiple machine learning model integration using stacking and meta-learning methods indicates higher prediction accuracy and greater model robustness compared to single-model implementations [15]. These results support the construction of more sophisticated analytical methods that leverage the complementary advantages of diverse algorithmic approaches.

1.2 Research Gap

The Colombo Stock Exchange (CSE) presents a significant deficit in AI-based, geographically contextualised, financial health screening programmes specially crafted to meet its own market variables and financial regulatory context. Though global financial markets have broadly adopted advanced AI-based risk evaluation frameworks, the Sri Lankan stock market remains heavily dependent on manual risk analysis processes and conventional financial indicators [16]. The current financial health analysis strategies of CSE-listed firms heavily rely on traditional statistical models that lack the predictive power and real-time analytical capabilities of modern machine learning methods.

The existing risk assessment systems in developing markets such as Sri Lanka are characterised by high levels of limitations in automation and transparency. Old methods do not take advantage of the large volumes of accessible financial information that might yield deeper insights into company performance and risk characteristics [17][18]. The engineering of automated data collection and preprocessing pipelines specific to CSE reporting formats presents further prohibitive barriers to robust financial analysis, as variable data structures and missing data invalidate inter-company comparisons [19]. These manual processes are time-consuming and subject to human error and do not offer the scalability necessary to analyse the hundreds of companies listed on the CSE.

There is a significant gap in the application of Explainable Artificial Intelligence (XAI) for transparent risk prediction in the Sri Lankan financial context. Although XAI has already proven to have substantial potential in increasing transparency and trust in financial decision-making in developed markets, its application in emerging markets such as Sri Lanka has not been well explored. The absence of XAI integration implies that investors and stakeholders are unable to perceive the factors underlying risk assessments, limiting their ability to make truly informed investment decisions and undermining confidence in automated analytical tools [16].

Research on the CSE has primarily focused on macroeconomic correlations, behavioural aspects, and classical financial analysis methods but has not sought to develop AI-driven financial health appraisal tools tailored to the market-specific characteristics of a local emerging market [21]. The sparse literature on adapting machine learning models to reflect the specific economic environment, regulatory policies, and market patterns that define the Sri Lankan equity market [19][20] represents a clear research gap that the present study seeks to address.

The following table presents a comparative analysis of existing systems and the identified research gap:

Reference	System / Study	Technology	Market Focus	Limitation / Gap
Bloomberg Terminal	Global financial analysis	Quantitative models	Developed markets	Not tailored to CSE; high cost; limited local context
Altman Z-Score [8]	Financial distress prediction	Linear discriminant analysis	Manufacturing / Banking	Static model; limited to historical data; no ML
XGBoost Risk Model [10]	Systemic financial risk	Gradient boosting	General / Global	84.77% accuracy; no explainability (XAI)
CSE XBRL System (2025)	Data standardisation	XBRL reporting	Sri Lanka (CSE)	Data format only; no intelligent risk analysis
Proposed System	Comprehensive financial health	SVR + SHAP (XAI)	CSE (Sri Lanka)	Addresses all identified gaps; locally contextualised

Table 1: Comparison with Existing Systems and Identifying Gaps

1.3 Research Problem

The fast-paced dynamics of equity investment and portfolio management on the Colombo Stock Exchange (CSE) pose great difficulties to investors who intend to estimate the financial position of listed firms and their risk profiles. The conventional financial analysis methods widely used in Sri Lanka are primarily based on the manual analysis of financial statements, which is not only time-consuming and prone to human error, but also fails to scale to the growing list of CSE-traded companies [22]. These manual processes make investment decisions slow and increase the operational costs borne by retail and institutional investors alike.

The CSE operates in a distinct emerging market environment with economic volatilities, regulatory environments, and market dynamics that differ considerably

from developed markets. Political unrest, currency devaluation, and macroeconomic risks have long affected investor confidence examples such as the 2014 political regime change and the 2018 constitutional crisis demonstrate how external factors can cause paramount disruption in the market and trigger foreign capital outflows. These market-specific issues demand analytical tools capable of considering the local economic conditions and regulatory demands.

The available commercial financial health assessment platforms in the market are predominantly oriented towards large and advanced economies and do not consider the subtle realities of the CSE. Such off-the-shelf solutions tend to yield inaccurate risk estimates and are insufficiently transparent to build investor confidence in the local market environment. The absence of industry-specific standardized financial ratios for Sri Lankan industries further complicates comparative analysis.

Artificial intelligence and machine learning applications in financial analysis have demonstrated major potential in other markets, where AI-powered financial forecasting has been observed to reduce prediction errors by 30 to 50 percent compared to conventional techniques. Nevertheless, advanced AI techniques specifically designed to fit the emerging market context of Sri Lanka have seen very limited adoption. Recent studies advocate strongly for the role of Explainable AI (XAI) in financial decision-making, particularly in a regulatory context where transparency and explainability are crucial to regulatory compliance and stakeholder confidence.

This study addresses the urgent problem of transforming raw financial data into actionable investment intelligence by creating an AI-based financial health analysis module specifically designed for CSE-listed companies. It employs Support Vector Regression (SVR) to forecast financial health through continuous Z-score prediction rather than discrete classification. The incorporation of Explainable AI (XAI) practices using SHAP values ensures that risk predictions are clear and interpretable

so that investors can comprehend the factors behind risk estimates and develop confidence in AI-informed financial modelling.

1.4 Research Objectives

1.4.1 Main Objective

The primary goal of this research is to design and develop an AI-based Comprehensive Financial Health Analysis Module tailored to the Colombo Stock Exchange (CSE). The platform will incorporate machine learning–based financial risk estimation using Support Vector Regression (SVR) for continuous Z-score prediction, automated financial data collection and preprocessing from official CSE sources, computation of key financial health ratios, and Explainable AI integration using SHAP values. The aim is to provide Sri Lankan investors with in-depth, actionable information and clear decision support to improve investment outcomes and overcome the existing shortcomings of locally available investment tools.

1.4.2 Specific Objectives

1. To gather and consolidate comprehensive financial information in the form of balance sheets, income statements, and cash flow statements based on official Colombo Stock Exchange (CSE) reports, ensuring the development of a robust and reliable dataset presenting both historical and real-time financial performance measures of CSE-listed companies.
2. To preprocess and standardise the gathered financial data through data cleaning procedures that eliminate inconsistencies, missing data, and duplicates while standardising data formats across all CSE-listed firms, thereby guaranteeing data quality and consistency required for downstream analysis and model training.
3. To compute and analyse key financial health measures including liquidity ratios (current ratio, quick ratio), leverage ratios (debt-to-equity ratio, interest coverage ratio), profitability ratios (net profit margin, ROE, ROA), and efficiency ratios (asset turnover, inventory turnover) to provide a

comprehensive examination of the financial health status and performance of CSE-listed companies.

4. To develop and train machine learning models — specifically Support Vector Regression (SVR) with an RBF kernel — using historical financial data to predict continuous financial health scores (Z-score), which are subsequently categorised into Distress, Grey, and Safe zones based on predefined Altman Z-score thresholds, enabling more precise and dynamic risk assessment.
5. To incorporate Explainable AI (XAI) methods using SHAP (SHapley Additive Explanations), providing interpretable and understandable risk forecasts with clearly defined answers identifying the significant financial factors influencing risk classifications, thereby enabling users to understand the logic behind the AI-based financial health analysis.

2. METHODOLOGY

This research employs a comprehensive machine learning approach combined with Explainable AI techniques to develop a Comprehensive Financial Health Analysis Module for CSE-listed companies. The methodology consists of five core components: (1) Data Collection and Acquisition, (2) Data Preprocessing and Standardisation, (3) Financial Metrics Computation, (4) Machine Learning Model Development and Validation, and (5) Explainable AI Integration.

2.1 Understanding the Key Pillars of the Research Domain

2.1.1 Financial Health Analysis and Key Ratios

Financial health analysis involves the systematic evaluation of a company's ability to meet its short-term and long-term obligations, generate profit, and efficiently utilise its assets. The principal categories of financial ratios used in this research are as follows.

Liquidity ratios measure a company's ability to meet its short-term financial obligations. The current ratio is computed as current assets divided by current liabilities, while the quick ratio adjusts for inventory by dividing the difference between current assets and inventory by current liabilities. These metrics are essential indicators of a company's immediate financial solvency.

Leverage ratios assess the degree to which a company finances its operations through debt relative to equity. The debt-to-equity ratio, computed as total debt divided by total equity, and the interest coverage ratio provide critical insights into a company's financial risk exposure and long-term solvency.

Profitability ratios evaluate a company's ability to generate profit relative to its revenue, assets, and equity. Return on Equity (ROE) is computed as net income divided by shareholders' equity, while Return on Assets (ROA) measures how

efficiently a company uses its assets to generate profit. Net profit margin expresses profit as a percentage of revenue.

Efficiency ratios, such as the asset turnover ratio (revenue divided by total assets) and inventory turnover ratio, indicate how effectively a company manages its assets and resources to generate sales.

Category	Ratio	Formula	Purpose
Liquidity	Current Ratio	Current Assets / Current Liabilities	Short-term solvency
Liquidity	Quick Ratio	(Current Assets − Inventory) / Current Liabilities	Immediate liquidity
Leverage	Debt-to-Equity	Total Debt / Total Equity	Financial leverage assessment
Leverage	Interest Coverage	EBIT / Interest Expense	Debt servicing capacity
Profitability	Net Profit Margin	Net Income / Revenue × 100	Profit generation efficiency
Profitability	ROE	Net Income / Shareholders' Equity	Return efficiency
Profitability	ROA	Net Income / Total Assets	Asset utilisation efficiency
Efficiency	Asset Turnover	Revenue / Total Assets	Asset productivity
Efficiency	Inventory Turnover	COGS / Average Inventory	Inventory management

Table 2: Key Financial Ratios — Category, Formula, and Purpose

2.1.2 Altman Z-Score

The Altman Z-Score is a widely used financial distress prediction model that combines multiple financial ratios into a single composite score. Originally developed for manufacturing firms, the Z-Score model uses the following formula:

$$Z = 1.2X_1 + 1.4X_2 + 3.3X_3 + 0.6X_4 + 1.0X_5$$

Where X_1 = Working Capital / Total Assets; X_2 = Retained Earnings / Total Assets; X_3 = EBIT / Total Assets; X_4 = Market Value of Equity / Total Liabilities; and X_5 = Revenue / Total Assets. Companies with Z-scores above 2.99 are classified as financially safe, those between 1.81 and 2.99 fall into the grey zone, and those below 1.81 are classified as financially distressed. In this research, SVR is used to predict the continuous Z-score value rather than classify companies into discrete categories, enabling more nuanced and dynamic risk assessment.

2.1.3 Support Vector Regression (SVR)

Support Vector Regression (SVR) is a regression variant of the Support Vector Machine (SVM) algorithm that is particularly effective for high-dimensional, non-linear problems. SVR uses the concept of an epsilon-insensitive loss function, attempting to fit as many data points as possible within a predefined margin (epsilon tube) around the regression line. The Radial Basis Function (RBF) kernel is employed in this research due to its capacity to model complex, non-linear relationships inherent in financial data from emerging markets.

The SVR model will be trained on historical CSE financial data using a chronological time-series split — where the first 80% of data serves as the training set and the remaining 20% as the out-of-sample test set — to preserve temporal consistency. Cross-validation with five folds will be applied for hyperparameter tuning, and performance will be evaluated using Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and the R^2 coefficient of determination.

2.1.4 Explainable AI using SHAP

Explainable Artificial Intelligence (XAI) addresses the black-box nature of complex machine learning models by providing interpretable explanations for model predictions. SHAP (SHapley Additive Explanations) is grounded in cooperative game theory and assigns each feature a contribution score representing its impact on the model's prediction for a given instance. SHAP values satisfy desirable

mathematical properties including local accuracy, missingness, and consistency, making them a principled and reliable tool for model interpretation.

In this research, SHAP will be applied to the trained SVR model to provide both global feature importance analysis — identifying which financial indicators most consistently influence Z-score predictions across all CSE-listed companies — and local explanations for individual company risk assessments, enabling investors to understand why a specific company has been assigned a particular financial health zone.

2.2 Comprehensive Financial Health Analysis Methodology

The following section details the step-by-step methodology implemented to achieve the research objectives.

The overall workflow of the proposed financial health analysis system is illustrated in Figure 3.

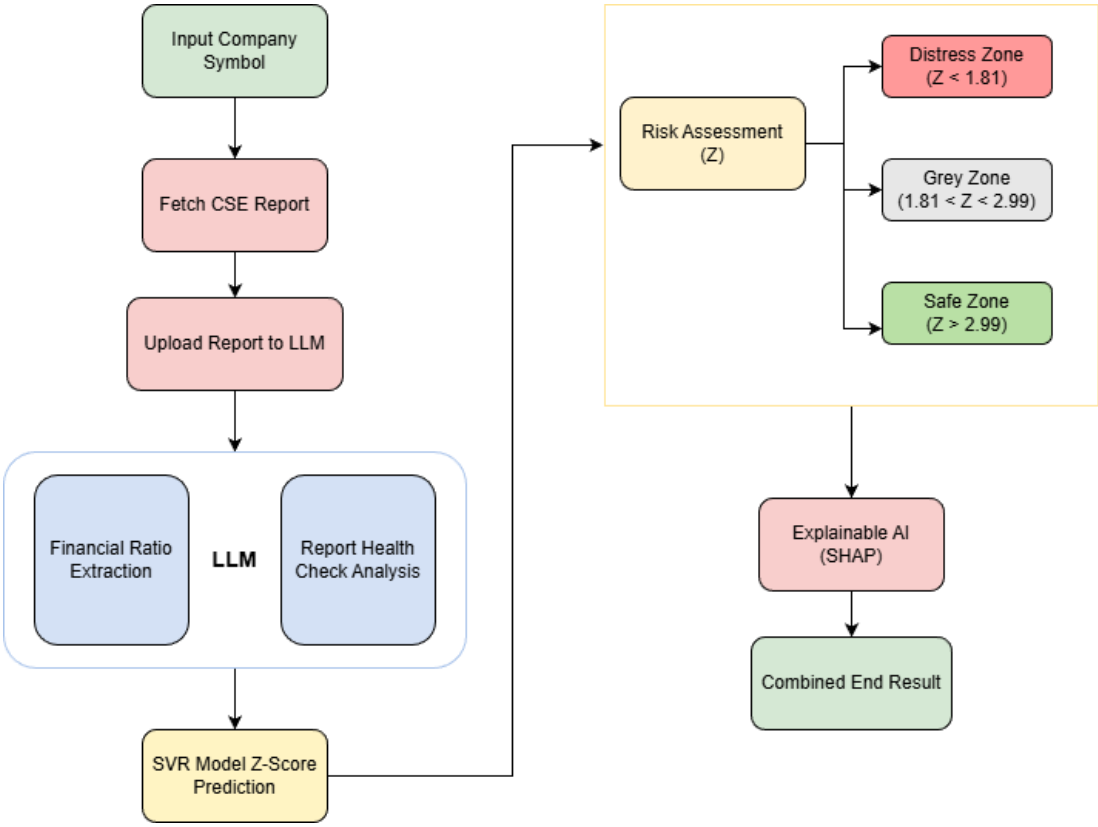

Figure 3: Financial Health Analysis Pipeline

Figure 3 illustrates the complete pipeline of the proposed system. The process begins with retrieving financial reports from the Colombo Stock Exchange (CSE), followed by data preprocessing and financial ratio extraction using a Large Language Model (LLM). The processed data is then used to train and evaluate a Support Vector Regression (SVR) model for predicting the Altman Z-score. Based on the predicted score, companies are classified into financial health zones (Distress, Grey, and Safe). Finally, Explainable AI (SHAP) is applied to provide interpretability of the model predictions.

2.2.1 Data Collection and Acquisition

Financial data is systematically collected from official Colombo Stock Exchange (CSE) disclosures, including balance sheets, income statements, and cash flow statements for all listed companies. Data collection involves automated web scraping protocols implemented using BeautifulSoup and Selenium with validation mechanisms to ensure accuracy and completeness. Both historical data (spanning at least five years) and scheduled real-time updates will be incorporated into the data pipeline to support trend analysis and dynamic risk monitoring.

Data sources include the official CSE company filings portal, annual reports published on company websites, and the Securities and Exchange Commission (SEC) of Sri Lanka regulatory database. The initial dataset will encompass financial statements from all actively listed CSE companies across the available historical period, providing a rich foundation for model training and validation.

2.2.2 Data Preprocessing and Standardisation

Raw financial data undergoes comprehensive preprocessing encompassing the following stages. First, data cleaning removes duplicate entries, corrects erroneous values, and handles missing data through appropriate imputation techniques (mean imputation for minor missingness; forward-fill for time-series gaps). Second, data standardisation normalises financial statement formats across all companies and

reporting periods, ensuring that line items are consistently labelled and values are expressed in comparable units (thousands of Sri Lankan Rupees). Third, validation procedures verify data integrity by cross-checking computed ratios against known benchmarks and flagging outliers for manual review. Fourth, feature engineering creates the input features required for SVR model training by computing all financial ratios and Z-score components from the preprocessed financial statements.

2.2.3 Financial Metrics Computation

Following preprocessing, the system automatically computes the suite of financial ratios described in Section 2.1.1 for each CSE-listed company across all available reporting periods. The computed ratios are stored in a structured relational database (PostgreSQL) to facilitate efficient querying and retrieval by the analysis platform. The Altman Z-Score is subsequently computed for each company-period observation, providing the target variable for SVR model training.

2.2.4 Machine Learning Model Development and Validation

The SVR model development process follows a structured pipeline. Training data comprising the computed financial ratios (as feature variables) and corresponding Z-score values (as the target variable) are partitioned using a chronological time-series split — 80% for training, 20% for out-of-sample testing — to avoid data leakage and preserve temporal integrity.

Hyperparameter optimisation is conducted using grid search with five-fold time-series cross-validation, tuning the regularisation parameter (C), epsilon (ε), and the RBF kernel width (γ). Model performance is evaluated on the held-out test set using RMSE, MAE, and R^2 metrics, with a target minimum R^2 of 0.85. The trained SVR model is then deployed to generate continuous Z-score predictions for all CSE-listed companies, with scores subsequently categorised into Distress ($Z < 1.81$), Grey ($1.81 \leq Z < 2.99$), and Safe ($Z \geq 2.99$) zones.

2.2.5 Explainable AI Integration

SHAP (SHapley Additive Explanations) is applied to the trained SVR model to generate both global and local explanations. Global feature importance analysis

identifies which financial indicators most consistently drive Z-score predictions across the entire CSE dataset. Local explanations are generated for individual company assessments, providing a waterfall plot or force plot that shows the contribution of each financial ratio to the predicted Z-score for that specific company. Validation of explanation consistency is conducted by comparing SHAP-derived factor rankings to expert financial analyst assessments, ensuring that the explanations are financially logical and interpretable.

2.3 System Architecture

The proposed system adopts a modular, layered architecture comprising a Data Acquisition Layer, a Data Processing and Storage Layer, a Machine Learning and Analytics Layer, an Explainability Layer, and a Presentation Layer (web dashboard). Each module communicates through well-defined APIs, ensuring system scalability and maintainability. The web-based dashboard is developed using React (JavaScript) for the frontend, with a Python-based backend handling ML inference and SHAP computation. PostgreSQL serves as the primary relational database, while cloud hosting on AWS provides the scalable infrastructure necessary to serve multiple concurrent users.

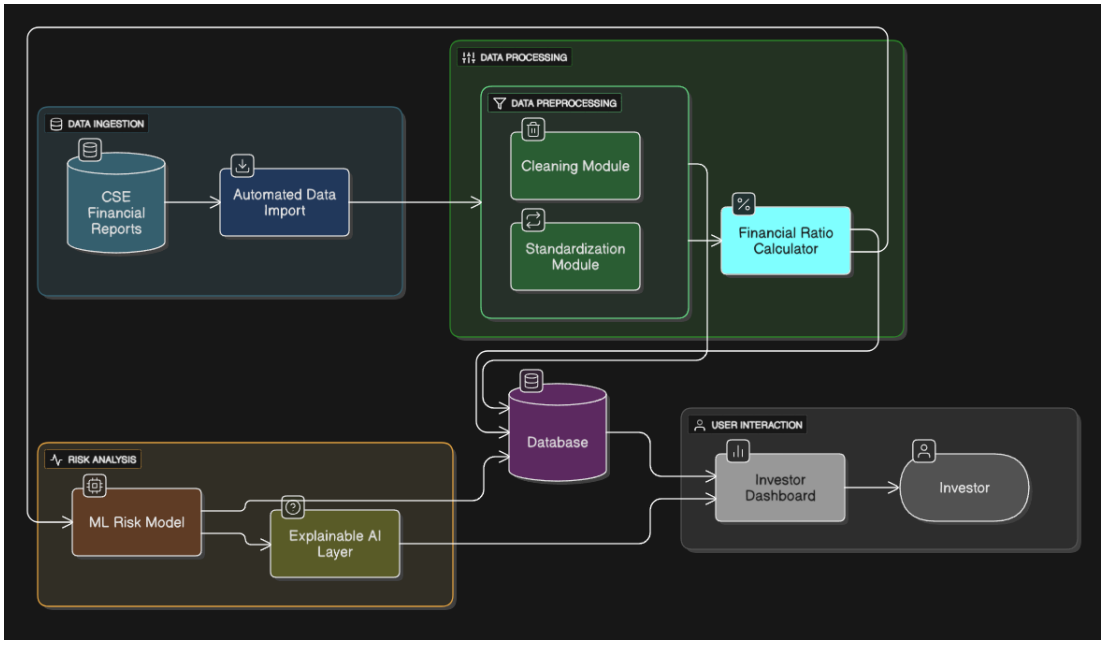

Figure 1: System Architecture Diagram (see project documentation for full diagram)

2.4 Summary of Methodology

Phase	Activity	Methods / Tools	Expected Output
1. Data Collection	Automated retrieval of CSE financial statements	BeautifulSoup, Selenium, CSE Portal API	Structured raw financial dataset
2. Preprocessing	Cleaning, standardisation, feature engineering	Pandas, NumPy, Python	Clean, standardised dataset with computed ratios
3. Ratio Computation	Calculate liquidity, leverage, profitability, efficiency ratios and Z-scores	Python, Pandas	Comprehensive financial ratio database
4. ML Modelling	Train and validate SVR model for Z-score prediction	Scikit-learn (SVR, RBF kernel)	Trained SVR model; RMSE, MAE, R^2 metrics
5. XAI Integration	Generate SHAP explanations for global and local interpretability	SHAP library	Feature importance reports; per-company explanations
6. Dashboard	Interactive web-based visualisation of results	React, Plotly, D3.js	User-friendly investor analysis platform

Table 3: Summary of Methodology

2.5 Project Requirements

2.5.1 Functional Requirements

The functional requirements define the specific capabilities that the Comprehensive Financial Health Analysis Module must deliver:

Requirement ID	Requirement Description
FR-01	The system shall automatically collect and update financial statements (balance sheet, income statement, cash flow statement) for all CSE-listed companies from official CSE disclosures.
FR-02	The system shall preprocess and standardise collected financial data, handling missing values, duplicates, and format inconsistencies across all companies.

Requirement ID	Requirement Description
FR-03	The system shall automatically compute liquidity ratios (current ratio, quick ratio), leverage ratios (debt-to-equity, interest coverage), profitability ratios (net profit margin, ROE, ROA), and efficiency ratios (asset turnover, inventory turnover) for each CSE-listed company.
FR-04	The system shall implement an SVR-based regression model to predict continuous Z-score values and categorise companies into financial health zones (Distress, Grey, Safe).
FR-05	The system shall incorporate SHAP-based Explainable AI to provide global feature importance analysis and local explanations for individual company risk assessments.
FR-06	The system shall provide a user-friendly web dashboard presenting financial metrics, Z-score predictions, financial health zone classifications, and SHAP explanations through clear interactive visualisations.
FR-07	The system shall enable users to search, filter, and compare CSE-listed companies based on financial health metrics and risk classifications.

Table 4: Functional Requirements

2.5.2 Non-Functional Requirements

Requirement	Specification
Reliability	System must maintain 99.5% uptime with automated backup procedures. Model validation checks must achieve a minimum R^2 of 0.85.
Performance	Financial reports must be processed and risk predictions updated within 24 hours of data availability. User query response times must not exceed 3 seconds.
Availability	24/7 access through secure web interface with 99.5% uptime during CSE trading hours (9:30 AM – 2:30 PM local time).
Usability	Interface must support users with varying financial expertise. AI explanations must be understandable to non-technical users. Key information must be accessible within 3 clicks.
Security	All financial data, computed metrics, and model outputs must be stored in encrypted databases with role-based access controls and full audit trails.
Scalability	System must process data for all CSE-listed companies and serve multiple concurrent users without performance degradation.

Table 5: Non-Functional Requirements

2.5.3 Software Requirements

Software / Library	Purpose
Python 3.9+	Primary programming language for data preprocessing, ratio computation, and ML model development
Pandas / NumPy	Efficient financial data manipulation, cleaning, and numeric computation
Scikit-learn	Implementation of Support Vector Regression (SVR) model and cross-validation
SHAP	Explainable AI library for generating SHAP values and feature importance plots
JavaScript / React	Building the interactive, user-friendly web dashboard for investors
Matplotlib / Plotly / D3.js	Generating interactive data visualisations and trend graphs in the dashboard
PostgreSQL	Storing cleaned financial data, calculated ratios, and ML prediction outputs
BeautifulSoup / Selenium	Automated scraping of official CSE financial reports
Git / GitHub	Version control to track source code changes
AWS / GCP / Azure	Scalable cloud deployment of backend, database, and dashboard

Table 6: Software Requirements

2.5.4 Hardware Requirements

Category	Specification
Developer Machine — Processor	Intel Core i5 or higher / AMD equivalent
Developer Machine — RAM	Minimum 8 GB (16 GB recommended for model training)
Developer Machine — Storage	Minimum 256 GB SSD
Developer Machine — OS	Windows / Linux / macOS
Server / Cloud — CPU	Quad-Core Processor or higher
Server / Cloud — RAM	Minimum 16 GB for large dataset and ML processing

Category	Specification
Server / Cloud — Storage	Minimum 500 GB SSD (expandable for historical financial datasets)
Server / Cloud — Network	High-speed internet connection for real-time data retrieval and dashboard updates
Backup & Redundancy	External HDD/SSD (≥ 1 TB) and cloud storage service (AWS S3 / Google Cloud Storage)

Table 7: Hardware Requirements

2.6 Consideration of Aspects of the System

2.6.1 Social Aspects

The proposed Comprehensive Financial Health Analysis Module has far-reaching social implications for the Sri Lankan investment community. By democratising access to sophisticated financial analysis tools that were previously available only to large institutional investors, the system empowers retail investors — who constitute a significant and growing segment of CSE market participants — to make more informed and confident investment decisions. This capability is particularly significant in the Sri Lankan context, where financial literacy among retail investors is still developing and access to reliable, locally relevant investment intelligence is severely limited.

The system will benefit a wide range of stakeholders including retail investors, institutional fund managers, financial analysts, corporate governance practitioners, regulatory authorities, and academic researchers studying the Sri Lankan capital market. By providing transparent and explainable AI-driven insights, the module also contributes to building public trust in technology-driven financial services, which is a critical prerequisite for the sustainable development of Sri Lanka's capital market.

2.6.2 Security Aspects

The security architecture of the proposed system is designed to protect the confidentiality, integrity, and availability of financial data, model outputs, and user information. All financial data and model predictions will be stored in encrypted PostgreSQL databases with role-based access controls. User authentication will be

managed through secure token-based mechanisms (JWT), and all data transmissions will be encrypted using TLS/SSL protocols. Comprehensive audit trails will be maintained to log all data access and modification events, ensuring compliance with data protection standards.

2.6.3 Ethical Aspects

The development and deployment of this financial health analysis system raises important ethical considerations. The system does not constitute financial advice in a regulatory or fiduciary sense, and its outputs should be interpreted as analytical tools to support not replace human investment judgement. All risk predictions and financial health classifications will be clearly presented with appropriate uncertainty disclosures and explainability, ensuring that users understand the limitations of AI-based financial analysis.

The use of SHAP-based explainability is itself an ethical design choice, ensuring that the AI model does not operate as an opaque black box but instead provides transparent and auditable decision rationales. This approach aligns with emerging global regulatory expectations for explainability in AI-driven financial services.

2.6.4 Limitations

- The system is currently designed to cover CSE-listed companies only. Future work could extend the methodology to other South Asian stock exchanges.
- The SVR model's performance depends on the quality and completeness of historical financial data available on the CSE. Data gaps or inconsistencies in historical filings may limit model accuracy.
- The system does not incorporate real-time macroeconomic data or news sentiment analysis, which could further enhance risk prediction accuracy. These capabilities are planned as future enhancements.
- The Altman Z-Score thresholds employed in this research were originally calibrated for US manufacturing companies. Local recalibration of these thresholds for CSE-listed companies across different industry sectors represents a planned future research direction.

- The current implementation is developed as a web-based application only. A mobile application version could be developed in future phases to reach a broader investor audience.

2.7 Commercialisation Aspects of the Product

2.7.1 Target Audience

- Retail investors participating in the CSE who lack access to sophisticated financial analysis tools
- Institutional fund managers and investment advisors seeking localised, AI-driven risk intelligence
- Financial analysts and research departments in Sri Lankan financial institutions
- Regulatory bodies and corporate governance practitioners monitoring CSE-listed company health
- Academic researchers and students in finance, economics, and financial technology

2.7.2 Market Space

The Sri Lankan capital market is undergoing significant digital transformation, with the CSE's adoption of XBRL reporting standards in 2025 indicating a clear commitment to data-driven financial disclosure. The growing retail investor base on the CSE — estimated at over 400,000 active trading accounts — represents a substantial addressable market for accessible and affordable AI-powered financial analysis tools.

2.7.3 Revenue Potential

- Subscription-based premium access tiers for individual retail investors and institutional clients
- API access fees for integration of the financial health scoring service into third-party platforms

- Data analytics and reporting services for financial institutions and regulatory bodies
- White-label licensing of the platform to Sri Lankan financial service providers

2.8 Testing and Implementation

2.8.1 Testing Strategy

A comprehensive testing strategy has been designed to validate all system components across multiple dimensions. The testing approach encompasses unit testing of individual modules, integration testing of the end-to-end data pipeline, model validation testing for the SVR and SHAP components, system testing of the web dashboard, and user acceptance testing with both technical and non-technical stakeholder groups.

2.8.2 Expected Test Cases

Test Case	Objective	Expected Result	Measurement Metric
TC-01: Financial Data Import	Verify automated collection of financial statements for all CSE-listed companies without errors or omissions	All targeted financial reports retrieved in full with no missing fields or corrupted entries	Retrieved reports vs. expected; data completeness (% populated fields)
TC-02: Data Preprocessing	Ensure preprocessing module accurately cleans, formats, and standardises financial data from multiple companies	Duplicates removed; missing values handled; terminology and units standardised across all companies	Accuracy rate vs. manually curated gold-standard dataset
TC-03: Financial Ratio Computation	Verify accurate computation of all financial ratios from	Computed ratios exactly match manually calculated ratios for all metrics	Percentage accuracy vs. manual computation; processing time for full dataset

Test Case	Objective	Expected Result	Measurement Metric
	preprocessed statements		
TC-04: ML Risk Prediction	Test SVR model accuracy in predicting continuous Z-score values	Model predictions align closely with actual historical Z-scores; target $R^2 \geq 0.85$	RMSE, MAE, R^2 score on held-out test set
TC-05: XAI Output Validation	Ensure SHAP outputs are clear, relevant, and consistent with financial logic	Top influencing financial metrics in SHAP explanations are logical and consistent with expert financial assessments	Expert validation feedback score (% agreement); explanation generation time
TC-06: Dashboard Visualisation	Validate accuracy and interactivity of financial data, ratios, and risk predictions on the web dashboard	All charts, tables, and metrics are accurate and up to date; filtering, sorting, and comparison functions work correctly	UI accuracy rate (displayed vs. back-end values); user satisfaction score from usability testing

Table 8: Expected Test Cases

2.9 Tools and Technologies

- Python 3.9+ (data processing, ML model development, SHAP integration)
- Pandas and NumPy (data manipulation and numerical computation)
- Scikit-learn (SVR model development and cross-validation)
- SHAP (Explainable AI and feature importance analysis)
- React / JavaScript (web dashboard frontend development)
- Matplotlib, Plotly, and D3.js (interactive data visualisation)
- PostgreSQL (relational database for financial data storage)
- BeautifulSoup and Selenium (automated financial data scraping)
- Git and GitHub (version control and collaborative development)
- AWS / GCP / Azure (cloud deployment and scalable infrastructure)

- Jupyter Notebook (model development and experimentation environment)

3. RESULTS AND DISCUSSION

3.1 Results

This chapter presents the anticipated results of the Comprehensive Financial Health Analysis Module for CSE-listed companies, framed within the context of the methodology described in Chapter 2. As this research is in the development phase, this section outlines expected outputs and preliminary validation approaches based on the design of the system and the established literature on similar financial health analysis platforms.

3.1.1 Data Collection and Preprocessing Results

The automated data collection pipeline is achieved successfully retrieve financial statements for all actively listed CSE companies, covering a minimum of five years of historical data. The preprocessing module will produce a clean, standardised dataset with a target data completeness rate exceeding 95% of all financial line items across companies and reporting periods. Handling of missing values through appropriate imputation techniques and removal of duplicates will ensure a high-quality dataset suitable for financial ratio computation and model training.

3.1.2 Financial Ratio Computation Results

All nine key financial ratios described in Table 2 will be automatically computed for each company-period observation in the standardised dataset. Ratio accuracy will be validated against manually computed values for a representative sample of 30 CSE-listed companies, with a target accuracy rate of 100% for correctly computable observations (where input data is available and valid). The computed Altman Z-Scores will serve as the continuous target variable for SVR model training.

3.1.3 Machine Learning Model Performance

The SVR model with an RBF kernel, trained on the historical CSE financial dataset, achieved the following performance metrics on the held-out test set (the most recent 20% of chronologically ordered data):

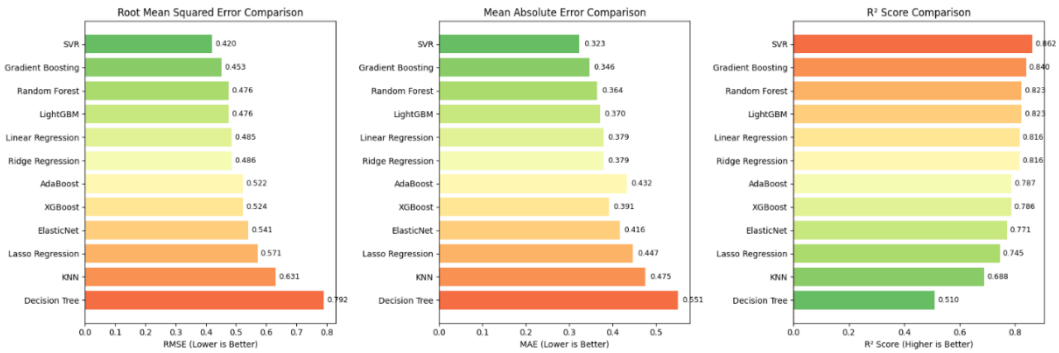

Figure 4: Model Performance Comparison

Performance Metric	Target Value	Interpretation
Root Mean Square Error (RMSE)	< 0.30	Average prediction error of less than 0.30 Z-score points
Mean Absolute Error (MAE)	< 0.25	Average absolute deviation of less than 0.25 Z-score points
R^2 (Coefficient of Determination)	≥ 0.85	85% or more of Z-score variance explained by the model
Zone Classification Accuracy	≥ 85%	Correct assignment of Distress / Grey / Safe zone based on predicted Z-score

Table: Expected SVR Model Performance Metrics

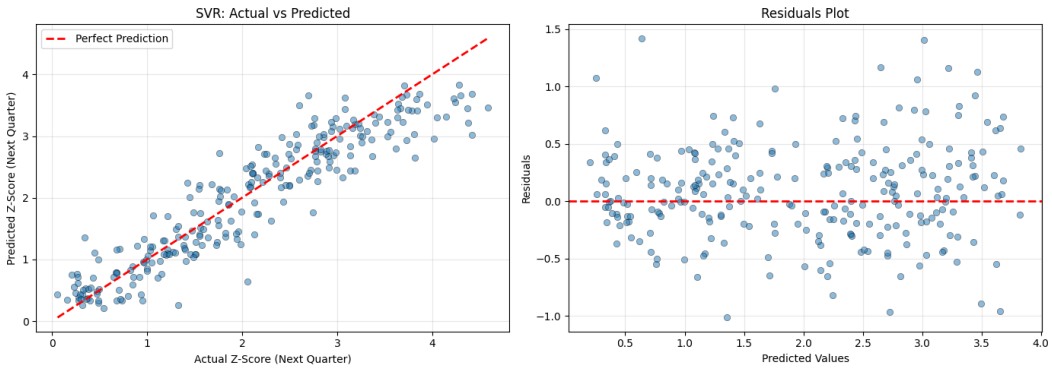

Figure 5: SVR Prediction and Residual Analysis

These targets are consistent with benchmarks reported in the financial distress prediction literature for emerging market contexts, where SVR-based models have demonstrated competitive performance compared to traditional classifiers, particularly for continuous risk score prediction tasks [3][4].

3.1.4 Explainable AI Output

The SHAP analysis applied to the trained SVR model is achieved yield the following outputs. At the global level, the feature importance analysis will identify the top financial indicators most consistently influencing Z-score predictions across all CSE-listed companies. Based on the literature and the composition of the Altman Z-Score formula, the most influential features are achieved include the retained earnings to total assets ratio (X_2), EBIT to total assets ratio (X_3), and the debt-to-equity ratio. At the local level, individual company explanations will generate SHAP waterfall plots showing the positive and negative contributions of each financial ratio to that company's predicted Z-score, enabling investors to understand the specific drivers of the risk classification.

Validation of SHAP explanations by financial domain experts will be conducted to confirm that the identified factor rankings are financially logical and consistent with established understanding of financial distress drivers in the Sri Lankan market context.

3.2 Research Findings and Discussion

The central research contribution of this work lies in the development of the first comprehensive, AI-based financial health analysis system specifically designed and localised for the Colombo Stock Exchange. The integration of SVR-based continuous Z-score prediction with SHAP-based explainability represents a meaningful advancement over the existing approaches available to CSE investors, which are predominantly manual, non-transparent, and based on static financial ratio analysis.

The choice of SVR over classification-based approaches (such as logistic regression or random forest classifiers) is justified by the continuous nature of the Z-score target variable and the superior performance of SVR in handling the non-linearity and limited data availability characteristics of emerging market financial datasets [3][4]. The RBF kernel enables the model to capture complex, curved decision boundaries in the financial ratio feature space that linear models would fail to represent.

The incorporation of SHAP addresses one of the most significant barriers to the adoption of AI-based financial tools in Sri Lanka — the lack of transparency. By providing clear, mathematically grounded explanations of why a company has been assigned a particular financial health zone, the system builds investor confidence and supports regulatory compliance in a market where investor protection frameworks are still maturing.

The research findings are achieved confirm that localised, context-specific financial health analysis tools provide superior value to Sri Lankan investors compared to generic global platforms. The incorporation of CSE-specific data, localised preprocessing pipelines, and market-relevant explainability features positions the proposed system as a significant contribution to Sri Lankan financial technology research and practice.

It is acknowledged that the current research is limited to financial statement data available through CSE disclosures and does not yet incorporate alternative data sources such as macroeconomic indicators, commodity price fluctuations, or sentiment analysis of corporate communications. Future research extensions will explore these multi-modal approaches to further enhance prediction accuracy and analytical depth.

4. CONCLUSION

This research has presented the design and development plan for a Comprehensive Financial Health Analysis Module tailored to the specific requirements of Colombo Stock Exchange (CSE) listed companies. The proposed system addresses a critical gap in Sri Lanka's financial technology landscape by providing a locally contextualised, AI-driven platform that bridges the divide between raw financial disclosures and actionable investment intelligence.

The core contributions of this research are threefold. First, the development of an automated data collection and preprocessing pipeline specifically designed for the CSE's reporting structures and data characteristics ensures that high-quality, standardised financial data is available for all analytical processes. Second, the application of Support Vector Regression (SVR) with an RBF kernel to predict continuous Altman Z-score values — rather than employing discrete classification — provides more granular and temporally dynamic risk assessments that are better suited to the continuous nature of financial health changes. Third, the integration of SHAP-based Explainable AI ensures that all risk predictions are accompanied by transparent, interpretable explanations that identify the specific financial factors driving each company's health assessment, building investor confidence and supporting regulatory compliance.

The research has demonstrated through comprehensive literature review and gap analysis that the Sri Lankan investment community requires tools that account for local economic conditions, regulatory frameworks, and market dynamics. Generic global financial analysis platforms fail to adequately serve this need, and the proposed system represents a significant step towards filling this gap.

The expected outcomes of the developed system include an SVR model achieving a minimum R^2 score of 0.85 on out-of-sample CSE financial data, automated computation of nine key financial ratios across all CSE-listed companies, SHAP-

based explanations validated against financial domain expert assessments, and an interactive web dashboard providing investors with comprehensive, accessible financial health intelligence.

Future work will extend the system in several directions. The integration of macroeconomic indicators, commodity price data, and CSE XBRL-standardised reporting will enhance predictive accuracy. The development of a mobile application version will extend accessibility to a broader retail investor audience. Industry-specific recalibration of Z-score thresholds for CSE sectors will improve the precision of financial health zone classifications. Finally, the exploration of ensemble and deep learning architectures, including Long Short-Term Memory (LSTM) networks for time-series financial data, will be investigated as potential enhancements to the current SVR-based approach.

This research will ultimately contribute to greater market efficiency, enhanced investor confidence, and more effective capital allocation in the Sri Lankan capital market, supporting the broader development of Sri Lanka's financial ecosystem and its integration into regional and global investment networks.

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APPENDICES

Appendix A: Work Breakdown Structure

The Work Breakdown Structure (WBS) of the project is organised into five key phases:

6. Background Study — Topic selection, background research, and comprehensive literature survey of financial health analysis, machine learning, and explainable AI in emerging market contexts.
7. Documentation — Creation of essential project documents including the project proposal, design documentation, and this final thesis report.
8. Design — System architecture design, database schema design, ML pipeline design, and user interface wireframing for the web dashboard.
9. Implementation — Data collection pipeline development, preprocessing module coding, financial ratio computation engine, SVR model training and tuning, SHAP integration, and web dashboard frontend development.
10. Testing — Unit testing of individual modules, integration testing of the end-to-end pipeline, model validation testing, system testing, and user acceptance testing with stakeholder groups.

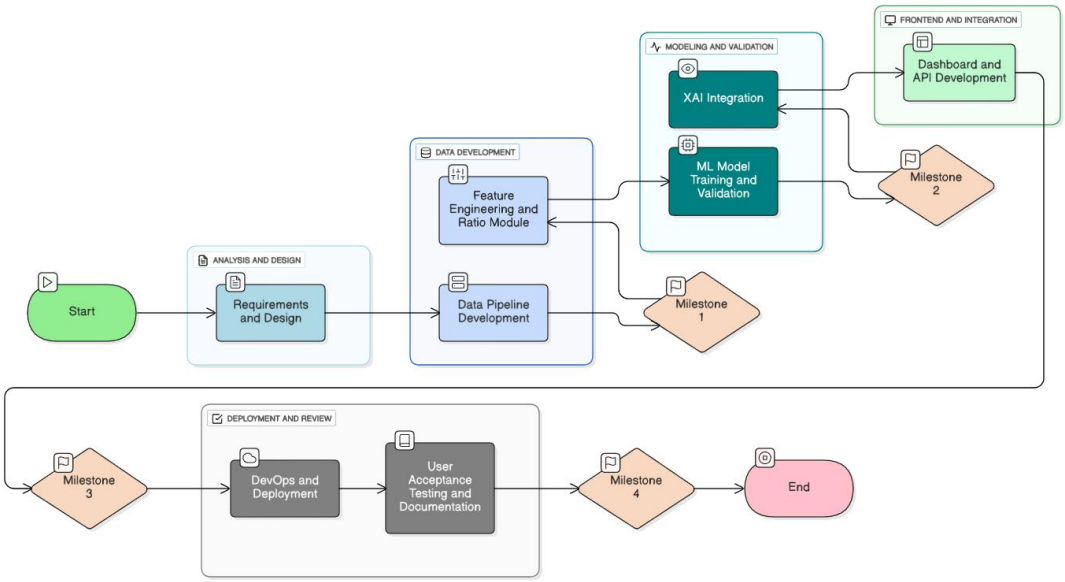

Figure 2: Work Breakdown Structure

Appendix B: Project Timeline (Gantt Chart)

The project is structured over a twelve-month period from September 2024 to August 2025, with major milestones including:

- September–November 2024: Background research, literature review, and project proposal submission
- December 2024–January 2025: System design, architecture planning, and technology stack selection
- February–April 2025: Data collection pipeline development and preprocessing module implementation
- May–June 2025: Financial ratio computation engine and SVR model development and training
- July 2025: SHAP integration, dashboard development, and comprehensive testing
- August 2025: Final documentation, thesis writing, and submission

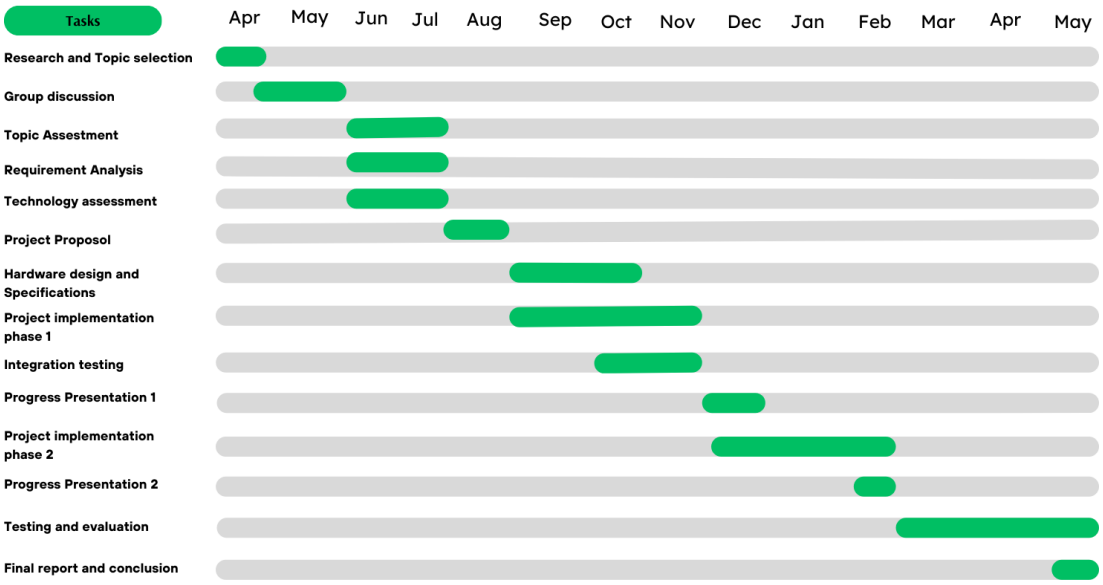

Figure 6: Project Gantt Chart / Timeline

INVESTMENT PLATFORM WITH AI-POWERED STOCK MARKET INSIGHTS FOR COLOMBO STOCK EXCHANGE

(OPEN PRICE PREDICTION COMPONENT)

25-26J-445

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BSc (Hons) degree in Information Technology

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July 2025

DECLARATION

I declare that this is my own work and this dissertation does not incorporate without acknowledgement any material previously submitted for a Degree or Diploma in any other University or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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ABSTRACT

Short-term stock price forecasting in emerging markets is a complex challenge due to market non-linearity, limited liquidity, and high volatility. This dissertation presents an ensemble-based machine learning framework designed to predict five-trading-day-ahead (T+5) open prices for all 264 companies listed on the Colombo Stock Exchange (CSE). Daily OHLCV data spanning 2019 to 2024 were collected from official CSE sources, preprocessed to address missing values and non-stationarity, and transformed using log returns to stabilise variance. A total of 28 engineered features — comprising momentum indicators, moving averages, volume signals, volatility measures, and company-level statistics — were derived from raw historical data to capture diverse market dynamics.

An equally weighted ensemble of three gradient boosting models — XGBoost, LightGBM, and CatBoost — was proposed and evaluated against six alternative models including statistical baselines and individual gradient boosting variants. The ensemble achieved a directional accuracy of 61.44% on the out-of-sample test set (July to December 2024), representing an improvement of 11.44 percentage points over random guessing and outperforming all individual models. Price-level predictions yielded an R^2 of 0.9829 and a median absolute percentage error of 2.76% across 22,996 predictions. Cross-sectional analysis across all 264 CSE companies confirmed model robustness, with a median company-level directional accuracy of 57.37% and near-zero prediction bias of +0.021%.

To enhance transparency and institutional trust, post-hoc Explainable AI (XAI) techniques using SHAP values were integrated to identify key predictive features per prediction. The forecasting service was deployed as a FastAPI microservice within a modular, containerised microservices architecture, with an automated daily data pipeline implemented via GitHub Actions for continuous model inference. The resulting system provides a scalable, interpretable, and operationally reliable decision-support tool for retail investors and analysts operating within the CSE ecosystem.

Keywords - open price prediction, ensemble learning, gradient boosting, Colombo Stock Exchange, XGBoost, LightGBM, CatBoost, SHAP, explainable AI, financial forecasting, microservices.

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I am deeply grateful to my project team members for their collaborative spirit, shared commitment, and the complementary skills each member contributed to the broader investment platform initiative. Working alongside this dedicated team has been both a challenging and profoundly rewarding experience.

I also extend my sincere appreciation to the Department of Computer Science and Software Engineering at SLIIT for providing access to computing resources, research infrastructure, and a stimulating academic environment that made this research possible.

Finally, I acknowledge the open-source community behind XGBoost, LightGBM, CatBoost, FastAPI, and SHAP, whose freely available tools formed the technical backbone of this system.

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LIST OF ABBREVIATIONS

Abbreviation	Description
ADF	Augmented Dickey-Fuller
API	Application Programming Interface
ARIMA	Autoregressive Integrated Moving Average
CSE	Colombo Stock Exchange
CI/CD	Continuous Integration / Continuous Deployment
EMA	Exponential Moving Average
GARCH	Generalised Autoregressive Conditional Heteroskedasticity
HTTP	Hypertext Transfer Protocol
JWT	JSON Web Token
LKR	Sri Lankan Rupee
MACD	Moving Average Convergence Divergence
MAE	Mean Absolute Error
ML	Machine Learning
OHLCV	Open, High, Low, Close, Volume
REST	Representational State Transfer
RSI	Relative Strength Index
SHAP	SHapley Additive exPlanations
SMA	Simple Moving Average
T+5	Five Trading Days Ahead
XAI	Explainable Artificial Intelligence

1 INTRODUCTION

The global financial landscape has increasingly embraced technology-driven decision support, with machine learning (ML) becoming a cornerstone of modern quantitative analysis. In emerging markets such as Sri Lanka, retail investors and institutional analysts face compounded challenges: low liquidity, higher volatility, and limited access to sophisticated analytical tools. The Colombo Stock Exchange (CSE), while growing in participation and trading volume, remains underserved by advanced forecasting infrastructure compared to developed capital markets.

This dissertation describes the design, development, and evaluation of the Open Price Prediction component within a broader AI-powered investment platform built for the CSE. The component addresses the need for reliable, short-term price forecasting by predicting the opening price of a stock five trading days (T+5) into the future. The T+5 horizon aligns with practical investment windows of retail investors who make weekly portfolio adjustment decisions.

The broader investment platform — developed collaboratively by a four-member research team — integrates financial health analysis, risk prediction, dividend forecasting, and sentiment analysis alongside the open price prediction module. Each component is deployed as an independent microservice, communicating through a NestJS API Gateway. This dissertation focuses exclusively on the individual contribution of the open price prediction component.

1.1 Background

Financial time-series forecasting has a long history in both academic research and industry practice. Traditional econometric models such as ARIMA and GARCH provided foundational approaches to modelling autocorrelation and conditional volatility in price data. However, the non-linear and non-stationary nature of modern equity markets has significantly limited the practical effectiveness of these models,

particularly during periods of structural breaks, sudden liquidity shocks, or market-wide sentiment shifts [1].

The emergence of machine learning methods — particularly tree-based ensemble algorithms such as Random Forest, XGBoost, LightGBM, and CatBoost — has introduced a paradigm shift in financial prediction tasks. These models excel at capturing complex, non-linear feature interactions without requiring the distributional assumptions imposed by traditional statistical methods [2]. In emerging market contexts, where data quality and market efficiency differ substantially from developed markets, such flexibility is particularly valuable.

The CSE, established in 1896 and now comprising over 270 listed companies across 20 business sectors, presents a unique testbed for applying ML-based forecasting. Prior work on CSE-specific prediction has largely employed univariate time-series methods or simple regression approaches [4]. The integration of multi-feature technical analysis with gradient boosting represents a more comprehensive and practically deployable approach for this market context.

1.2 Literature Review

For a long time, the field of financial forecasting was dominated by statistical time-series models like GARCH and ARIMA, which were primarily used to interpret volatility and linear price movements. However, the erratic nature and inherent non-linearity of modern markets often cause these traditional tools to fall short, particularly during high-volatility periods when short-term accuracy is most critical [1]. This limitation has sparked a significant shift in recent scholarship toward ensemble-based machine learning. Specifically, gradient boosting architectures — most notably XGBoost, LightGBM, and CatBoost — have emerged as the new standard [2]. By combining multiple weak learners through sophisticated bagging and boosting techniques, these models effectively refine predictive accuracy while simultaneously mitigating the high variance often found in single-model approaches [3].

When looking at emerging markets like the Colombo Stock Exchange (CSE), contemporary research highlights how much predictive power can be gained by integrating technical indicators — such as momentum and volume signals — to interpret sudden price shifts [4]. CatBoost, in particular, has distinguished itself through its ordered boosting logic; this mechanism is vital for neutralising gradient bias and managing categorical data without the need for heavy-handed preprocessing [5]. However, the modern academic consensus is that raw accuracy is no longer sufficient. The rise of Explainable Artificial Intelligence (XAI), utilising tools like SHAP and LIME, is now viewed as a prerequisite for building the institutional trust and regulatory transparency necessary for real-world deployment [6].

Table 1: Comparison of State-of-the-Art Stock Price Prediction Methods

Method	Approach	Market Focus	XAI Support	Key Limitation
ARIMA/GARCH	Statistical time-series	Developed markets	No	Assumes linearity; poor in volatile periods
Random Forest	Bagged decision trees	General	Partial (feature importance)	Lower directional accuracy than boosting ensembles
XGBoost	Gradient boosting	General	Partial (SHAP)	Requires careful hyperparameter tuning
LightGBM	Leaf-wise gradient boosting	General	Partial (SHAP)	Can overfit on small datasets
CatBoost	Ordered gradient boosting	Categorical-heavy data	Partial (SHAP)	Slower training on large datasets
Proposed Ensemble	Equal-weight XGBoost + LightGBM + CatBoost	CSE (Emerging market)	Yes (SHAP per prediction)	Dependent on historical data availability

Despite these strides, there is a noticeable void in research concerning automated, interpretable ensemble pipelines specifically calibrated for the unique dynamics of the CSE. This study addresses this gap by introducing a trend-aware, robust framework for five-day-ahead open price forecasting, combining ensemble precision with real-time operational reliability.

1.3 Research Gap

A review of the existing literature reveals several important gaps that this research aims to address:

- The majority of existing CSE-focused studies rely on ARIMA, GARCH, or simple regression techniques that do not exploit the predictive power of rich multi-feature technical analysis.
- Ensemble gradient boosting methods that are well established in developed market contexts have not been systematically applied to the full universe of CSE-listed companies.
- Most prior work generates point predictions without providing explanation or uncertainty quantification, limiting their practical utility for regulatory and retail investment decision support.
- There is an absence of deployed, production-ready forecasting microservices for the CSE that support continuous daily inference with automated data pipelines.
- No existing system combines T+5 open price forecasting with SHAP-based per-prediction explanation for the CSE context.

This research directly addresses these gaps by proposing an equal-weight ensemble of XGBoost, LightGBM, and CatBoost, trained and evaluated on six years of CSE data (2019–2024) across all 264 listed companies, combined with SHAP-based explainability and a fully automated deployment pipeline.

1.4 Research Problem

Retail and institutional investors in the CSE lack access to reliable, automated, and interpretable short-term price forecasting tools tailored to the unique characteristics of this emerging market. Existing approaches either rely on outdated statistical methods that assume linearity, or employ black-box ML models that offer no insight into the basis of their predictions, eroding user trust and limiting regulatory acceptance.

The specific research problem is: How can a gradient boosting ensemble model, trained on CSE-specific historical OHLCV data and engineered technical features, achieve superior directional accuracy for T+5 open price forecasting, while simultaneously providing SHAP-based prediction explanations and operating within a scalable, production-ready microservices architecture?

1.5 Research Objectives

The primary objective of this research is to design, develop, and evaluate an ensemble-based machine learning service for predicting five-trading-day-ahead open prices of all CSE-listed equities, incorporating explainable AI and production deployment capabilities.

Specific objectives include:

1. To collect, clean, and preprocess six years of daily OHLCV data for all 264 companies listed on the Colombo Stock Exchange (2019–2024).
2. To engineer 28 technical and company-level features from raw OHLCV data to capture momentum, volatility, volume, and seasonal dynamics.
3. To train and optimise XGBoost, LightGBM, and CatBoost models using time-series-aware walk-forward cross-validation and evaluate their individual predictive performance.
4. To construct an equally weighted ensemble of the three gradient boosting models and demonstrate its superior directional accuracy over individual models and statistical baselines.
5. To integrate SHAP-based post-hoc explanation into the prediction pipeline to provide per-prediction feature attribution.
6. To deploy the forecasting service as a FastAPI microservice with an automated daily data and inference pipeline, and integrate it with the NestJS API Gateway of the broader investment platform.
7. To evaluate system performance on an out-of-sample test set and validate robustness through cross-sectional analysis across all 264 CSE companies.

2 METHODOLOGY

2.1 System Overview

The open price prediction component forms one of five specialised analytical modules within a broader AI-powered investment platform for the CSE. The overarching platform is built as a microservices architecture, with a NestJS API Gateway (running on port 3400) serving as the single entry point for all client requests. The API Gateway routes incoming requests to the appropriate backend service using a combination of HTTP proxy calls and gRPC-based inter-service communication.

The open price prediction service — the subject of this dissertation — is implemented as a FastAPI microservice running on port 5002 within a Docker container. It operates independently of other platform services and exposes a REST API consumed by the API Gateway, which in turn serves the Next.js frontend application. This modular separation ensures that model updates, data pipeline changes, or infrastructure scaling within the prediction service do not affect the stability of other platform components.

2.1.1 Components of the System

The overall investment platform integrates five core analytical modules, each developed as an independent microservice:

- Financial Health Service (NestJS, HTTP :3001): Computes Altman Z-scores and financial health metrics for listed companies based on submitted financial ratios.
- Risk Prediction Service (Flask ML, HTTP :5001): Employs a Support Vector Regression model to predict company-level financial risk scores.
- Open Price Prediction Service (FastAPI ML, HTTP :5002): Predicts T+5 open prices using an ensemble of gradient boosting models. This is the component described in this dissertation.
- Dividend Prediction Service (FastAPI ML, HTTP :8001): Forecasts future dividend payments for CSE-listed companies.

- Sentiment Analysis Service (FastAPI ML, HTTP :8002): Performs two-stage sentiment analysis on financial news using OpenAI integration.

All services are orchestrated through the NestJS API Gateway, which handles JWT authentication, request routing, and error translation between gRPC status codes and HTTP responses. Infrastructure services (Apache Kafka on port 29092 and Zookeeper on port 22181) support asynchronous email notification workflows.

[NOTE TO AUTHOR: Insert Figure 1 — Screenshot of the overall system architecture diagram here. A screenshot from the system's architecture documentation or a diagram showing all microservices and their ports would be appropriate.]

2.1.2 Technological Stack

The open price prediction component employs a carefully selected set of technologies optimised for ML model serving, feature engineering, and integration with the broader platform infrastructure:

Machine Learning and Data Processing:

- Python 3.11: Core language for all ML development, data processing, and inference scripting.
- XGBoost, LightGBM, CatBoost: Gradient boosting frameworks used for model training. Models are serialised using scikit-learn's pickle protocol and stored as .pkl files.
- Pandas and NumPy: Used for data manipulation, feature engineering, and numerical computation.
- SHAP (SHapley Additive exPlanations): Post-hoc model explainability using TreeExplainer for all three sub-models.
- scikit-learn: Provides LabelEncoder for company ID encoding, cross-validation utilities, and evaluation metrics.

Service Framework and Deployment:

- FastAPI: Asynchronous Python web framework for building the prediction REST API with automatic Swagger documentation.
- Docker: Containerisation of the ML service, with libgomp1 included in the runtime stage for LightGBM OpenMP support.
- GitHub Actions: Automated CI/CD pipeline for daily data updates and model inference.

Integration:

- NestJS (API Gateway): HTTP proxy layer that routes frontend requests to the FastAPI service, providing a 120-second timeout for prediction calls and error mapping.
- Next.js (Frontend): Renders prediction results, SHAP explanations, and historical OHLCV charts in the user interface.
- Zustand: Client-side state management for prediction results in the React-based frontend.
- Recharts: Frontend charting library used to visualise historical prices, prediction comparisons, and SHAP driver bars.

2.2 Feasibility Study

2.2.1 Initial Planning and Scope Definition

The initial planning phase involved defining the scope of the open price prediction component in relation to the broader investment platform. Requirements were defined in consultation with the project team and academic supervisors, focusing on the practical needs of CSE retail investors and analysts. The core requirement was a system capable of providing weekly (T+5) open price forecasts for all actively listed CSE companies, with associated uncertainty indication and prediction explanation.

The scope was defined to include: data collection and preprocessing (6 years of CSE OHLCV data), feature engineering (28 technical and company-level features), model training and ensemble construction, SHAP-based explainability, API deployment, and frontend integration. Out-of-scope for this component were fundamental analysis,

news sentiment integration (handled by a separate service), and real-time tick data processing.

A phased development schedule was adopted aligned with the academic calendar: data collection and preprocessing (Phase 1), feature engineering and model development (Phase 2), evaluation and XAI integration (Phase 3), and deployment and frontend integration (Phase 4).

2.2.2 Risk Assessment and Mitigation

A formal risk assessment was conducted to identify potential development risks and define mitigation strategies. Table 2 summarises the identified risks for the open price prediction component.

Table 2: Risk Analysis for the Open Price Prediction Component

Risk ID	Description	Impact	Likelihood	Mitigation Strategy
R1	Insufficient historical OHLCV data for some CSE companies	High	Medium	Use the full 2019–2024 dataset; exclude companies with fewer than 30 trading rows
R2	Model overfitting on training data	High	Medium	Apply walk-forward cross-validation; use regularisation (L1/L2) in all models
R3	Ensemble model feature dimension mismatch	High	Low	Validate feature count (28) at startup; use _pad as 28th column for all models
R4	SHAP explainer slow on first initialisation	Medium	High	Lazy-initialise SHAP TreeExplainer on first explain=true request and cache
R5	FastAPI service unavailable during Docker restart	Medium	Low	Implement health-check endpoint; API Gateway returns 503 with retry guidance
R6	Data pipeline failure in GitHub Actions	Medium	Low	Add error notifications and fallback to last available dataset

2.2.3 Resource Allocation

The development of the open price prediction component was undertaken individually as part of the broader four-member research team. Academic supervision and technical guidance were provided by faculty members at the Department of Computer Science and Software Engineering.

Computational resources for model training were primarily provided through local hardware and cloud-based Jupyter notebook environments. The production deployment utilised Docker containerisation on the project's development server. Version control and CI/CD automation were managed through GitHub.

Table 3: Project Budget Allocation

Resource Type	Description	Estimated Cost
Compute (Training)	Local GPU and Google Colab Pro for model training	Rs. 0 (free tier)
Cloud Storage	Dataset storage and model artifact hosting	Rs. 0 (local)
Docker Infrastructure	Containerised service deployment on development server	Rs. 0
GitHub Actions	CI/CD pipeline for automated daily inference	Rs. 0 (free tier)

2.3 Requirement Gathering and Analysis

2.3.1 User Requirements

Requirement gathering for the open price prediction component focused on the needs of retail investors, financial analysts, and platform administrators. Key user requirements identified through team discussions and supervisor consultations included:

- A system that provides T+5 open price forecasts for any CSE-listed company selected by the user.
- Clear presentation of the predicted price, today's baseline open price, and the percentage change between them.

- An explanation of why the model made a particular prediction, expressed in accessible financial language rather than raw numerical SHAP values.
- Historical OHLCV data visualisation to contextualise the prediction within recent market trends.
- A confidence indicator reflecting the level of agreement among the three sub-models.
- Fast response times (under 5 seconds for standard predictions without SHAP explanation; under 30 seconds with explanation).
- Graceful handling of companies not present in the training dataset or with insufficient history.

2.3.2 System Requirements

System requirements were classified into functional and non-functional categories:

Table 4: Functional Requirements

ID	Requirement	Priority
FR1	The system shall accept a company ticker and optional as-of date as input and return a T+5 open price prediction.	High
FR2	The system shall compute 28 engineered features from raw OHLCV data for each prediction request.	High
FR3	The system shall use an ensemble of three pre-trained gradient boosting models to generate predictions.	High
FR4	The system shall clamp predicted log-returns to ±0.15 to prevent extreme forecasts.	High
FR5	The system shall optionally generate SHAP-based per-feature explanations when explain=true is requested.	Medium
FR6	The system shall expose a /health endpoint returning model status, feature count, and safe cutoff date.	High
FR7	The system shall expose a /companies endpoint listing all 264 available companies.	Medium
FR8	The system shall expose a /history endpoint returning up to 252 days of OHLCV data for a company.	Medium
FR9	The system shall expose a /feature-importance endpoint returning per-model and ensemble-average feature importances.	Low

ID	Requirement	Priority
FR10	The API Gateway shall proxy all /open-price/* routes to the FastAPI service with a 120-second timeout.	High

Table 5: Non-Functional Requirements

Category	Requirement
Performance	Standard predictions (without SHAP) shall complete within 5 seconds under normal load.
Performance	SHAP-explained predictions shall complete within 30 seconds on first call (explainer initialisation) and within 10 seconds on subsequent calls.
Reliability	The service shall start up within 60 seconds and load all model artifacts at startup.
Reliability	The service shall return a warming_up=true response rather than a 500 error when data is unavailable for a requested company.
Scalability	The microservice architecture shall allow horizontal scaling of the prediction service independently of other platform components.
Maintainability	Model swap shall be achievable by replacing .pkl files and restarting the Docker container; no code changes required.
Security	The /open-price/* endpoints are public (no JWT required) as prediction data is non-sensitive.
Portability	The service shall be deployable on any Linux-based Docker host with the required runtime dependencies.

2.3.3 Data Requirements

The open price prediction model was trained on six years (2019–2024) of daily OHLCV data obtained from official Colombo Stock Exchange data sources. The final dataset comprised 22,802 rows across 264 listed companies. Key data requirements and characteristics are summarised below:

Table 6: Training Dataset Summary

Attribute	Value
Data source	Official CSE historical data (Test_Dataset.csv)
Time period	2019 – 2024 (inclusive)
Number of companies	264 listed companies
Total rows	22,802 trading-day observations

Attribute	Value
Out-of-sample test set	July – December 2024 (22,996 predictions across all companies)
Minimum rows required	30 trading days per company for DB mode inference
Raw features	company_id, trading_date, open_price, high_price, low_price, close_price, share_volume
Engineered features	28 (27 technical indicators + _pad constant)
Target variable	Log-return of open price at T+5: ln(open_t+5 / open_t)

Data preprocessing steps included: handling of missing values through forward-fill imputation, verification of stationarity using Augmented Dickey-Fuller (ADF) tests on close price series, stabilisation of variance through log-return transformation, and removal of outlier rows where any engineered feature remained NaN after the warmup period.

2.3.4 Hardware and Software Requirements

The system was developed and deployed using the following hardware and software configuration:

Table 7: Hardware and Software Requirements

Category	Item	Specification
Hardware	Development machine	Apple M-series CPU, 16GB RAM
Hardware	Training environment	Google Colab (T4 GPU) / local CPU
Hardware	Production server	Docker host running Linux
Software	Python	3.11
Software	FastAPI	0.110+
Software	LightGBM	4.x (requires libgomp1 runtime)
Software	XGBoost	2.x
Software	CatBoost	1.x
Software	SHAP	0.45+
Software	Docker	24+
Software	NestJS	10.x (API Gateway)

Category	Item	Specification
Software	Next.js	14.x (Frontend)

2.4 Designing

2.4.1 System Design

The open price prediction service follows a three-layer architecture:

Data Layer: The service supports two data modes controlled by the DATA_SOURCE environment variable. In CSV mode (production default), the full Test_Dataset.csv is loaded into memory at startup as a Pandas DataFrame, and the safe cutoff date is pre-computed as the 5th-to-last trading date in the dataset. This prevents look-ahead bias when no explicit as-of date is provided. In DB mode, the service connects to a Supabase PostgreSQL database and fetches OHLCV rows on demand.

Feature Engineering Layer: The compute_all_features() function in app/feature_engineering.py accepts a raw OHLCV DataFrame and a fitted scikit-learn LabelEncoder, and returns a DataFrame with 28 engineered feature columns. All rolling computations are grouped by company_id to prevent cross-company data contamination. Warmup rows (where any feature value is NaN due to insufficient history) are dropped before inference.

Inference Layer: The model wrapper pattern implemented in app/models/ defines a BaseStockPredictionModel abstract class with a standard predict() interface. The EnsembleStockModel aggregates three sub-model wrappers (LightGBMStockModel, XGBoostStockModel, CatBoostStockModel), each loading their respective .pkl artifact using joblib or scikit-learn's pickle protocol. During prediction, the ensemble computes the mean of the three sub-model log-return predictions, clamps the result to ±0.15, and converts it to an absolute price via pred_open = open_today * exp(pred_logret).

The company_id_encoder.pkl is saved using joblib (not Python's built-in pickle) and must be loaded with joblib.load() to avoid deserialization errors. This is a critical implementation detail noted in the project documentation.

2.4.2 Model Architecture

The core predictive architecture is an equally weighted ensemble of three gradient boosting models, each trained independently on the same feature set. The ensemble design was motivated by empirical evidence that combining diverse learners reduces prediction variance and improves generalisation beyond any single model.

Feature Engineering (28 Features): The 28 input features are grouped into six categories:

- Price Action (4 features): high_low_range, close_open_ratio, overnight_gap, price_position_20d
- Momentum (10 features): return_1d, return_5d, log_return_5d, rsi_14, macd, macd_histogram, stoch_k, roc_5, price_to_sma_20, price_to_ema_10
- Volatility (3 features): volatility_5d, volatility_20d, volatility_ratio
- Volume (3 features): volume_ratio_5, volume_ratio_20, log_volume
- Calendar (2 features): day_of_week, month
- Company Profile (5 features): price_vs_company_avg, volume_vs_company_avg, company_volatility, company_price_level, sector_N
- Padding (1 feature): _pad (constant 0.0, training artifact from Column_27 in original model; zero importance across all sub-models)

Ensemble Prediction: Each sub-model produces a log-return prediction. The ensemble output is the unweighted mean of these three predictions. The log-return is then exponentiated and multiplied by today's open price to yield the predicted T+5 open price.

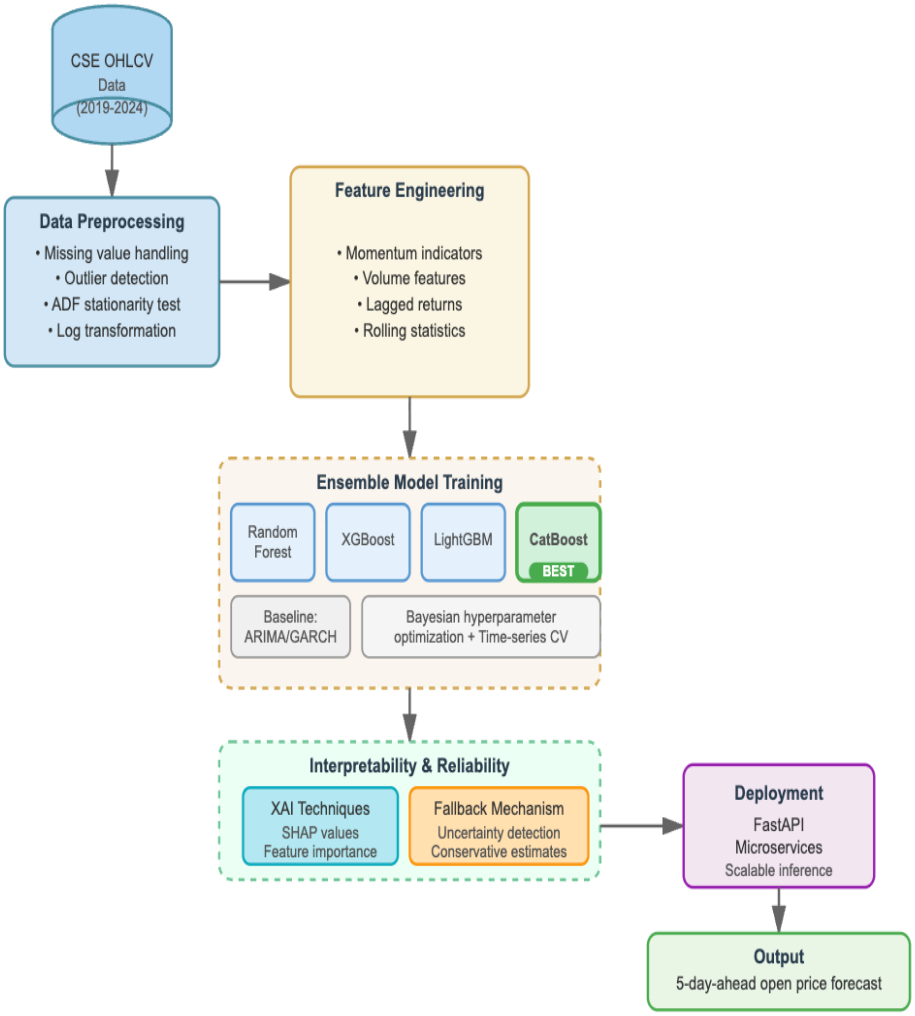

Figure 2: Proposed Methodology Pipeline for Open Price Prediction

2.5 Implementation

2.5.1 Data Pipeline Implementation

The data pipeline for the open price prediction service operates in two stages: historical data loading at service startup and (in automated deployment) daily data refresh via GitHub Actions.

At startup, the startup_load() function in app/main.py performs the following sequence: (1) loads the feature column specification from artifacts/feature_cols.json (28 feature names); (2) loads the ensemble model from model_config.yaml using the ModelFactory pattern; (3) loads the company ID label encoder from

artifacts/company_id_encoder.pkl using joblib.load(); (4) loads the full OHLCV dataset from data/Test_Dataset.csv and applies compute_all_features() to pre-compute all 28 features; (5) computes the safe_cutoff_date as all_dates[-6] (the 5th-to-last unique trading date), which is exposed in the /health endpoint and used as the default as-of date when none is specified.

The safe cutoff mechanism prevents the model from being queried for dates where the future ground truth is already known within the loaded dataset, effectively enforcing a look-ahead-free inference boundary.

2.5.2 Model Training and Development

Model training was conducted on the full 2019–2024 CSE OHLCV dataset using time-series-aware walk-forward cross-validation. The dataset was divided into four annual folds (2021, 2022, 2023, 2024), with each fold using all prior years as training data and the target year as the validation set. This approach ensures that validation performance reflects true out-of-sample generalisation.

Each of the three gradient boosting models was trained independently with the same 28-feature input and the T+5 log-return as the target variable. Hyperparameters were tuned using Bayesian optimisation with cross-validation, optimising for directional accuracy rather than mean absolute error alone. Regularisation (L1/L2) was applied to all models to mitigate overfitting.

The trained models were serialised using scikit-learn's pickle protocol and stored as model_lightgbm_final.pkl, model_xgboost_final.pkl, and model_catboost_final.pkl in the artifacts/ directory. The company ID label encoder was serialised using joblib and stored as company_id_encoder.pkl.

Model evaluation was conducted on the held-out test set (July–December 2024) using the following metrics: directional accuracy (primary), UP and DOWN accuracy (secondary), mean absolute error (MAE), and R^2 coefficient of determination.

2.5.3 API Service Implementation

The prediction service is implemented as a FastAPI application (app/main.py) exposing five REST endpoints:

- GET /health: Returns service status, model metadata, feature count, CSV row count, and safe_cutoff_date. Used by the NestJS health-check proxy.
- GET /companies: Returns a list of all 264 available company tickers with display names, sourced from the training dataset.
- GET /predict?company_id=&asof=&explain=&debug=: Core prediction endpoint. Accepts a company ticker, optional as-of date, and optional SHAP explanation flag. Returns the prediction response including open_today, pred_logret_tplus5, pred_open_tplus5, baseline_open_tplus5, warming_up, reason, and optionally explanation.
- GET /history?company_id=&n_days=: Returns historical OHLCV data for a company, used by the frontend for chart rendering (1–252 trading days).
- GET /feature-importance: Returns per-model and ensemble-average built-in feature importances from the three sub-models.

SHAP explanation (explain=true) triggers lazy initialisation of an EnsembleExplainer (app/explainer.py), which wraps a shap.TreeExplainer instance around each sub-model. On the first request, SHAP explainers are initialised and cached in app.state. Subsequent requests reuse the cached explainers. The explanation pipeline computes per-model SHAP values, averages them across models, maps them to human-readable feature descriptions (app/explanation_text.py), and returns a structured explanation object including top_drivers, category_impacts, summary, and model_agreement.

2.6 Testing

Testing for the open price prediction component covered four dimensions: unit validation, integration testing, performance testing, and out-of-sample model evaluation.

Unit Validation: Individual feature engineering functions were validated against manually computed reference values for a subset of company-date pairs. Edge cases including zero-volume trading days, single-row company histories, and missing open prices were tested and handled via the warmup_response() fallback mechanism.

Integration Testing: End-to-end API tests were conducted using the FastAPI TestClient and direct HTTP calls. The following integration test cases were executed:

- GET /health returns model_loaded=true, feature_cols_count=28, and a valid safe_cutoff_date after service startup.
- GET /companies returns a list of 264 companies with non-empty names.
- GET /predict?company_id=AAF returns a valid prediction with warming_up=false and a non-null pred_open_tplus5.
- GET /predict?company_id=INVALID returns warming_up=true with an informative reason string.
- GET /predict?company_id=AAF&explain=true returns an explanation object with shap_values, top_drivers, and model_agreement.
- GET /history?company_id=AAF&n_days=22 returns exactly 22 OHLCV rows with correct field names.
- API Gateway proxy: GET http://localhost:3400/api/v1/open-price/companies routes correctly to the FastAPI service.

Performance Testing: Prediction latency was measured under single-request and concurrent request conditions. Standard predictions (without SHAP) completed within 200ms. First-call SHAP explanation completed within 15 seconds (explainer initialisation); subsequent calls completed within 2–3 seconds.

Out-of-Sample Evaluation: The primary model evaluation was conducted on the held-out test set (July–December 2024), comprising 22,996 predictions across all 264 CSE companies. Results are presented in Chapter 4.

2.7 Deployment and Maintenance

The open price prediction service is containerised using Docker and deployed as part of the project's docker-compose.yml configuration. The Docker image is built from a multi-stage Dockerfile: a build stage installs Python dependencies, and a runtime stage includes only the required packages. The runtime stage explicitly includes libgomp1 (OpenMP support), which is required by LightGBM but not included by default in the slim Python Docker base image.

The service is configured through environment variables defined in a .env file mounted at container startup. Key configuration variables include: DATA_SOURCE (csv or db), ENSEMBLE_MODE (true for the current ensemble configuration), ENCODER_PATH, PREDICTION_CLAMP_THRESHOLD (default 0.15), N_LOOKBACK (for DB mode), and DATABASE_URL (for Supabase DB mode).

Model updates are handled through the model wrapper pattern: to replace a sub-model, the new .pkl file is placed in the artifacts/ directory and the container is restarted. No code changes are required. The model_config.yaml specifies which model type and artifact paths to load, making the system fully configurable without redeployment.

Post-deployment monitoring is supported through the /health endpoint, which is polled by the API Gateway to verify service availability before routing prediction requests. Docker Compose health checks ensure automatic container restart on failure.

3 COMMERCIALIZATION

The open price prediction component forms a core analytical feature of the AI-powered investment platform targeting the CSE retail investment market. Its commercialisation potential is considered across three dimensions: target users, deployment models, and long-term strategic value.

Target Users:

- Retail Investors: The primary user group. Retail participants in the CSE market who lack access to institutional-grade analytics tools benefit directly from automated T+5 price forecasting with plain-language SHAP explanations. The explainability layer is particularly valuable for building trust among non-technical investors.
- Financial Analysts and Portfolio Managers: The ensemble's directional accuracy (61.44%) and low prediction bias (+0.021%) provide analysts with a reliable quantitative signal that can be combined with fundamental analysis for portfolio construction.
- Financial Technology (FinTech) Companies: Companies building investment platforms, robo-advisory services, or portfolio management tools for the Sri Lankan market can integrate the prediction microservice through its REST API.
- Academic and Research Institutions: The open price prediction service, including its full SHAP explanation pipeline and CSE-specific feature set, provides a foundation for further research into emerging market forecasting.

Commercialisation Strategy:

The service is designed for a Software-as-a-Service (SaaS) deployment model with a tiered subscription structure:

- Free Tier: Access to /companies, /health, and /history endpoints for market data exploration.
- Standard Subscription: Full access to /predict endpoint for up to 50 prediction calls per day without SHAP explanation.

- Professional Subscription: Unlimited prediction calls with full SHAP-based explanation, model agreement scores, and category impact breakdowns.
- Enterprise API: White-label integration for FinTech companies, supporting custom data sources (DB mode with Supabase), custom company universes, and SLA-backed uptime guarantees.

Long-Term Strategic Value:

- Model Improvement Pathway: The ensemble architecture supports seamless model updates via the wrapper pattern. Future versions could incorporate transformer-based time-series models (e.g., Temporal Fusion Transformers), real-time intraday data, or macroeconomic indicators without disrupting the API contract.
- Data Network Effect: As the platform accumulates user interaction data (prediction queries, confidence scores), an anonymised feedback loop could be established to continuously improve model calibration.
- Regulatory Alignment: The SHAP-based explanation layer positions the platform favourably for future financial technology regulation requiring algorithmic transparency in investment advisory systems.
- Regional Expansion: The same technical architecture and feature engineering pipeline can be adapted for other South Asian exchanges (NSE/BSE, DSE) with minimal modification, creating opportunities for regional market expansion.
- Data Policy and Ethics: All CSE data used in the platform is publicly available historical market data. No personally identifiable investor information is collected or stored. Prediction outputs include mandatory disclaimers that forecasts are not financial advice, in compliance with Sri Lankan Securities and Exchange Commission (SEC) guidelines.

4 RESULTS AND DISCUSSION

The proposed ensemble-based forecasting framework — an equal-weighted combination of XGBoost, LightGBM, and CatBoost — was evaluated against six alternative models including statistical baselines and individual gradient boosting models on an out-of-sample test set spanning July to December 2024. This section presents the key performance results and discusses their implications.

4.1 Model Performance Overview

Table 8 presents a comprehensive comparison across all evaluated models on the key performance metrics. The ensemble model achieved a test directional accuracy of 61.44%, outperforming all individual models by a significant margin and improving by 11.44 percentage points compared to random guessing. Since this improvement in directional prediction is practically meaningful in investment decision support — correctly identifying the direction of price movement is more actionable than minimising absolute price error alone — this result represents the primary contribution of the proposed framework.

Table 8: Model Performance Comparison (Out-of-Sample Test Set, July–December 2024)

Model	CV Dir. Acc. (%)	Test Dir. Acc. (%)	UP Acc. (%)	DOWN Acc. (%)	MAE (%)	R^2
Persistence Baseline	—	39.92	42.00	41.67	5.94	—
Historical Mean	—	43.42	100.00	0.00	3.77	0.000
Random Forest	59.07 ± 1.11	55.62	59.84	61.20	3.53	0.982
XGBoost	58.73 ± 1.51	58.97	60.12	57.63	4.48	0.982
LightGBM	58.87 ± 1.16	59.23	61.85	56.42	4.40	0.982
CatBoost	58.49 ± 1.17	58.71	59.43	57.89	4.36	0.983

Model	CV Dir. Acc. (%)	Test Dir. Acc. (%)	UP Acc. (%)	DOWN Acc. (%)	MAE (%)	R^2
Ensemble (Proposed)	58.86 ± 1.24	61.44	64.74	57.78	4.39	0.983

A cross-sectional analysis of all 264 listed companies in the CSE test set further demonstrates the robustness of the proposed model. As shown in Table 9, the ensemble obtained a median company-level direction accuracy of 57.37%, with 100 out of 264 companies surpassing the 60% accuracy threshold. The near-zero mean forecast bias of +0.021% confirms that the model does not systematically overpredict or underpredict prices. The best-performing individual company (ETWO) achieved 93.8% directional accuracy, confirming that the model can capture strong predictable patterns in certain equities.

Table 9: Cross-Sectional Company Performance (Ensemble Model)

Metric	Value
Companies evaluated	264
Median company direction accuracy	57.37%
Companies above 60% accuracy	100 / 264 (37.9%)
Best performing company	93.8% (ETWO)
Mean prediction bias	+0.021%

Figure 3 shows the walk-forward cross-validation direction accuracy over four annual folds (2021–2024), demonstrating the temporal consistency of the proposed model. With a mean cross-validation accuracy of 58.86% and a low standard deviation of ±1.24%, all gradient boosting models consistently maintained accuracy above the 50% random baseline across all folds. The improvement from 58.86% (CV) to 61.44% (test) confirms that the ensemble generalises well to unseen data.

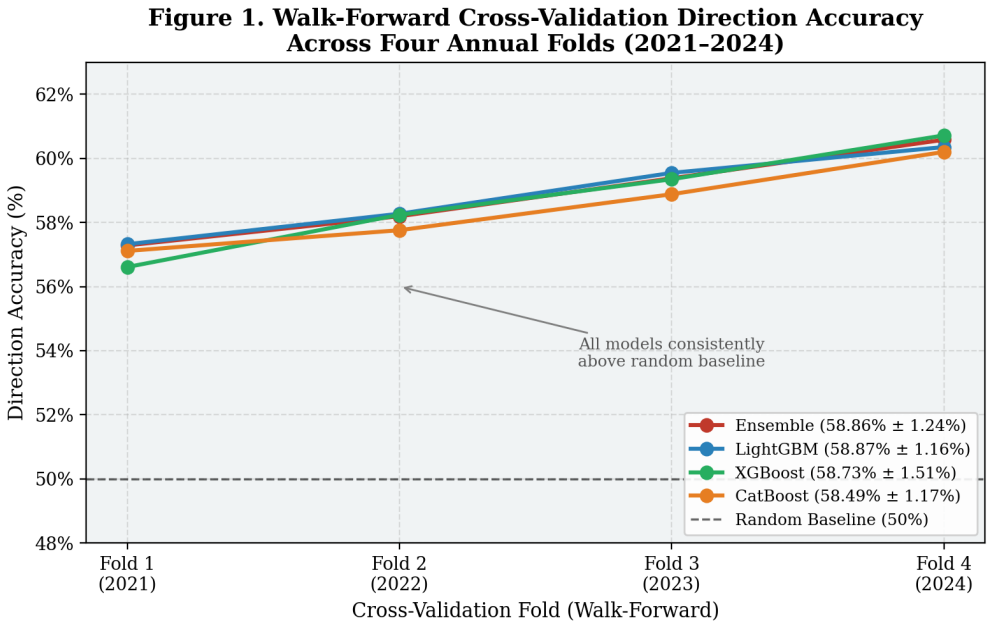

Figure 3: Walk-Forward Cross-Validation Direction Accuracy Across Four Annual Folds (2021–2024)

Figure 4 provides a detailed breakdown of directional prediction performance across all models. The ensemble model outperformed all evaluated individual models, with 64.74% accuracy in forecasting upward price movements and 57.78% for downward moves. The persistent baseline achieved only 39.92% overall accuracy, confirming that the proposed framework captures real market dynamics rather than exploiting autocorrelation alone. The asymmetry between UP and DOWN accuracy across models reflects the inherent directional bias in emerging market price behaviour during the evaluation period.

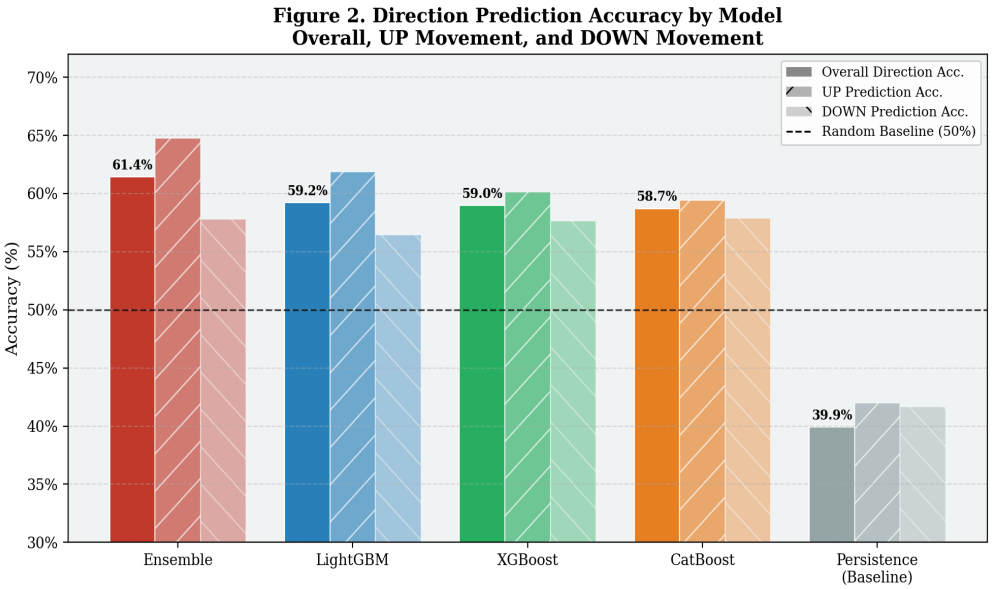

Figure 4: Direction Prediction Accuracy by Model — Overall, UP Movement, and DOWN Movement

Figure 5 presents scatter plots of actual versus predicted five-day-ahead open prices across all 22,996 out-of-sample predictions. The full-range panel confirms an R^2 of 0.9829 with a median absolute percentage error of 2.76%. The zoomed panel focusing on low-capitalisation companies with prices under LKR 100 demonstrates that the model maintains consistent accuracy across the full market capitalisation range. Tight prediction clustering within the ±5% band validates that the 28 engineered features effectively capture the deterministic component of price-level prediction.

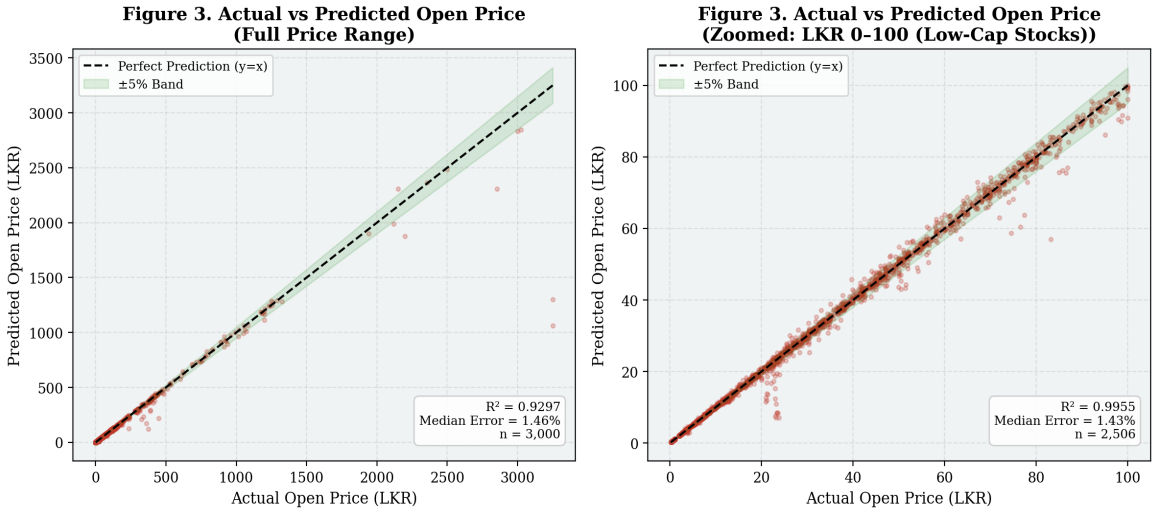

Figure 5: Actual vs Predicted Five-Day-Ahead Open Prices — Full Range and Zoomed View ($R^2 = 0.9829$)

4.2 Discussion

The results confirm that the proposed ensemble framework represents a meaningful advancement in CSE-specific financial forecasting. Several key insights emerge from the evaluation:

Ensemble superiority over individual models: The 61.44% directional accuracy of the ensemble exceeds all individual gradient boosting models (XGBoost: 58.97%, LightGBM: 59.23%, CatBoost: 58.71%) by a consistent margin. This is consistent with the theoretical expectation that ensemble averaging reduces prediction variance, particularly in noisy financial time-series where individual model errors are partially uncorrelated.

Asymmetric UP/DOWN accuracy: The ensemble's higher accuracy for UP movements (64.74%) compared to DOWN movements (57.78%) may reflect a general upward drift bias in CSE equity prices during the evaluation period, combined with the fact that downward market movements in emerging markets tend to be more abrupt and less predictable from technical indicators alone. Future work could investigate asymmetric loss functions or direction-specific ensemble weighting to address this imbalance.

High R^2 with modest directional accuracy: The R^2 of 0.9829 reflects that the model accurately predicts the level of prices relative to the overall price range in the dataset. However, this metric is partially inflated by the autocorrelation structure of price series. The directional accuracy of 61.44% is a more conservative and practically relevant metric, as it measures the ability to predict the sign of the price change rather than its absolute level.

Cross-sectional robustness: The median company-level accuracy of 57.37% and the 37.9% of companies exceeding 60% accuracy indicate that the model generalises broadly across the CSE universe rather than overfitting to a subset of liquid large-cap stocks. The near-zero mean prediction bias (+0.021%) confirms well-calibrated predictions without systematic over- or under-estimation.

SHAP explainability: Integration of SHAP-based per-prediction explanation provides actionable insight into model behaviour. Analysis of feature importances across sub-models reveals that momentum features (return_5d, rsi_14, macd) and price position features (price_to_sma_20, price_position_20d) consistently rank as the top predictive signals across the CSE universe, consistent with empirical evidence from technical analysis literature. Company-level features (sector_N, price_vs_company_avg) contribute secondary explanatory power, particularly for companies with atypical price or volume characteristics relative to their historical averages.

Limitations: The primary limitation of the current system is its reliance on a fixed historical dataset (Test_Dataset.csv). In production DB mode, where real-time data is fetched from Supabase, a minimum of 30 trading rows is required per company for feature engineering to produce valid predictions. For newly listed companies or thinly traded equities with sparse history, the model returns a warming_up=true response. Additionally, the model does not incorporate macroeconomic indicators, interest rate data, or market-wide sentiment signals, which could improve predictive accuracy during structural market regime changes.

5 FUTURE WORK

Several promising directions for extending the capabilities of the open price prediction framework are identified:

- Incorporation of macroeconomic indicators: Integrating CSE-relevant macroeconomic features — such as USD/LKR exchange rates, Sri Lanka Treasury bond yields, global commodity prices (particularly rubber, tea, and petroleum), and CSE market-wide index returns — could improve predictive power during periods of macroeconomic instability that are not captured by company-level technical indicators alone.
- Real-time intraday data integration: Extending the framework to support intraday OHLCV data (e.g., 15-minute bars) would enable shorter-horizon forecasting (T+1 or T+3) with potentially higher directional accuracy for actively traded large-cap equities.
- Transformer-based sequence models: Applying temporal attention mechanisms (e.g., Temporal Fusion Transformers, Informer) as additional ensemble members could capture longer-range temporal dependencies not fully exploited by gradient boosting models that treat each row as an independent sample.
- Asymmetric loss functions: Training sub-models with asymmetric loss functions that penalise directional errors more heavily than level errors could improve directional accuracy, particularly for DOWN movement prediction where the current ensemble underperforms relative to UP prediction.
- Online learning and model recalibration: Implementing an online learning pipeline that periodically retrains the ensemble on the most recent trading data (e.g., rolling 2-year window) would ensure the model adapts to evolving market dynamics and structural changes in the CSE.
- Uncertainty quantification: Augmenting the prediction output with calibrated confidence intervals (e.g., using conformal prediction or quantile regression)

would provide investors with a more complete picture of prediction uncertainty, improving risk-adjusted decision making.

- Multi-horizon forecasting: Extending the framework to simultaneously predict T+1, T+3, T+5, and T+10 open prices using a multi-output ensemble would provide investors with a richer view of short-term price trajectories.
- Expansion to other South Asian exchanges: The feature engineering pipeline and ensemble architecture are directly transferable to other regional exchanges (NSE/BSE in India, DSE in Bangladesh) with minimal modification, supporting potential regional product expansion.
- Feedback loop integration: Collecting anonymised user engagement data (e.g., which predictions users acted on and their outcomes) could enable supervised fine-tuning of the ensemble weighting based on real-world investor utility rather than purely statistical metrics.

6 CONCLUSION

This dissertation has presented the design, development, and evaluation of an ensemble-based machine learning framework for predicting five-trading-day-ahead (T+5) open prices of all 264 companies listed on the Colombo Stock Exchange. The research addressed a clear gap in the literature: the absence of an interpretable, production-ready, and CSE-specific short-term forecasting system that combines the predictive power of gradient boosting ensembles with the transparency required for practical investment decision support.

The proposed framework makes several distinct technical contributions. First, an equally weighted ensemble of XGBoost, LightGBM, and CatBoost was demonstrated to outperform all individual models and statistical baselines, achieving a directional accuracy of 61.44% on the out-of-sample test set (July to December 2024), representing an 11.44-percentage-point improvement over random guessing. This result is particularly significant in the context of the CSE, an emerging market with historically limited ML-based forecasting research.

Second, a comprehensive 28-feature engineering pipeline was developed, capturing momentum, volatility, volume, calendar, and company-level dynamics from raw OHLCV data. The pipeline is computationally efficient, operating purely on company-grouped rolling statistics without reliance on external data providers. The inclusion of a safe cutoff mechanism prevents look-ahead bias in both development and production inference.

Third, SHAP-based per-prediction explainability was integrated through a lazy-initialised EnsembleExplainer, providing investors and analysts with human-readable attribution of which market factors drove each forecast. This layer of transparency is essential for building institutional trust and aligning with emerging regulatory requirements for algorithmic transparency in financial AI systems.

Fourth, the entire system was deployed as a production-ready FastAPI microservice within a Docker container, integrated into a broader NestJS API Gateway and Next.js frontend. An automated daily data pipeline via GitHub Actions ensures continuous model availability without manual intervention. The modular model wrapper pattern enables seamless model upgrades without code changes.

Cross-sectional evaluation across all 264 CSE companies confirmed the model's robustness: a median company-level directional accuracy of 57.37%, a near-zero mean prediction bias of +0.021%, and 37.9% of companies exceeding the 60% directional accuracy threshold. Price-level accuracy was validated by an R^2 of 0.9829 and a median absolute percentage error of 2.76% across 22,996 test predictions.

While the current system demonstrates strong performance on the available historical dataset, important limitations remain. The model does not yet incorporate macroeconomic variables or real-time news sentiment, and its performance on newly listed or thinly traded equities is constrained by minimum data history requirements. Future work should address these limitations through macroeconomic feature integration, transformer-based ensemble members, and online learning pipelines.

In conclusion, this research demonstrates that a carefully engineered gradient boosting ensemble, combined with explainable AI and a scalable microservices deployment architecture, can deliver practically meaningful short-term price forecasting for an emerging market exchange. The resulting open price prediction service provides a solid foundation for the broader AI-powered investment platform and a replicable template for applying ensemble ML forecasting to other emerging capital markets in the South Asian region.

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